ADITYA UNIVERSITY

I B. Tech I/II SEMESTER

FRESHMAN ENGINEERING DEPARTMENT

BASIC ELECTRICAL & ELECTRONICS ENGINEERING LAB MANUAL
(241EE001)
ACADAMIC REGULATION: B. Tech – 2024

FACULTY MANUAL

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Vision

Aditya University aspires to be a globally recognized academic institution dedicated to quality education, cutting-edge research, and technological service to our country, and envisions itself as a beacon of holistic advancement and long-term impact, remaining dynamic in the ever-changing worlds of society, ecology, and economics.

Mission

- Aditya University pushes boundaries to design high-quality curricula and to provide students with a vibrant and relevant education that prepares them for a changing world. Our industry insights and creative teaching methods attempt to equip our students to be lifelong learners.
- Aditya University's learning environment encourages intellectual curiosity, critical thinking, and cooperation, with the goal of providing students with an immersive education that fosters creativity and innovation. Our cutting-edge facilities, interactive classrooms, and supportive faculty aim to motivate students to realize their full potential and contribute to society.
- Aditya University promotes cross-disciplinary inquiry and discovery and leads cutting-edge research and innovation. Through strategic partnerships, research grants, and a dedicated faculty, we aim to advance science, technology, and social sciences and empower students and faculty to conduct transformative research that solves real-world problems and elevates our institution globally.
- Aditya University is committed to producing world-changing business leaders and entrepreneurs through its emphasis on entrepreneurship, mentoring, and business incubation programs.

ADITYA UNIVERSITY

Program Outcomes (POs) - B.Tech

After successful completion of the program, the graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering Management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for

i) independent and life-long learning

ii) adaptability to new and emerging technologies and

iii) Critical thinking in the broadest context of technological change. (WK8)

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

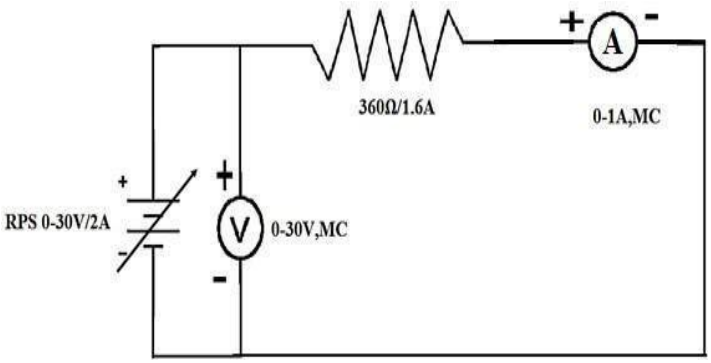
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OHM'S LAW

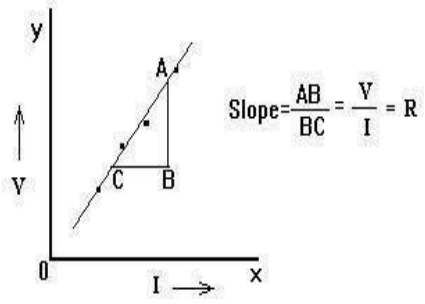
CIRCUIT DIAGRAM:



TABULAR FORM:

S. No.	Potential Difference V (volt)	Current I (A)	Resistance V / I = R (Ω)	Average Resistance R (Ω)
1				
2				
3				
4				
5				

GRAPH: Obtain value of `R` via slope of `V` vs `I `= _____Ω.



Experiment No:1

OHM'S LAW

AIM: To verify the ohm's law.

APPARATUS:

S.No.	Name of the Apparatus	Range	Type	Quantity
1	Regulated Power Supply	0-30V/2A	Transistorized	01
2	Ammeter	0-1A	MC	01
3	Voltmeter	0-30V	MC	01
4	Rheostat	360Ω/1.6A	WW	01
5	Connecting Wires			As Per Req.

THEORY:

Ohm's Law describes the relationship between voltage, current, and resistance in a circuit. It states that the current flowing through a conductor is directly proportional to the voltage across it, and inversely proportional to the resistance. Mathematically, this is expressed as V = IR, where V is voltage (in volts), I is current (in amperes), and R is resistance (in ohms).

OHM'S LAW LIMITATIONS:

1. Unilateral Devices:

Ohm's law doesn't hold for components like diodes and transistors, which allow current to flow in only one direction. The voltage-current relationship in these devices is not linear and depends on the direction of current flow.

2. Non-linear Elements:

Ohm's law is not applicable to components where the resistance is not constant but changes with voltage or current. Examples include capacitors, inductors, and non-ohmic materials like semiconductors (e.g., silicon and germanium).

3. Temperature Dependence:

The resistance of some materials (like wires) can change with temperature. Ohm's law, in its simplest form, assumes constant resistance, making it less accurate in situations where temperature varies significantly.

4. Frequency Dependence:

In AC circuits, the frequency of the alternating current can affect the resistance of a component, particularly for capacitors and inductors. Ohm's law is typically used for DC circuits, where frequency is not a factor.

5. Non-ideal Voltage Sources:

Ideal voltage sources provide a constant voltage regardless of the current drawn. Real-world voltage sources have internal resistance, which can affect the voltage drop across the circuit, making Ohm's law less accurate.

PROCEDURE:

1. The circuit is connected as per the circuit diagram.
2. Switch ON the Regulated Power Supply.
3. Vary The Voltage and Note down the Corresponding Ammeter reading.
4. Graph of V *Vs* I is drawn. A straight line is obtained.
5. The slope of the graph is found out. The slope gives the resistance (R) of the unknown resistance.

PRECAUTIONS:

1. All the connections must be very tight.
2. Record the current at the regular interval of voltage.
3. Least count of voltmeter and ammeter must be properly calculated.
4. Do not pass a large current through the resistance.
5. While measuring the voltage and current the needle of meters should not move out of the scale.

OBSERVATIONS:

1. Range of Voltmeter =_____V
2. Range of Ammeter =_____A
3. Least count of voltmeter (1Div)= _____V
4. Least count of ammeter (1Div)= _____A
5. E.M.F of the Battery = _____V

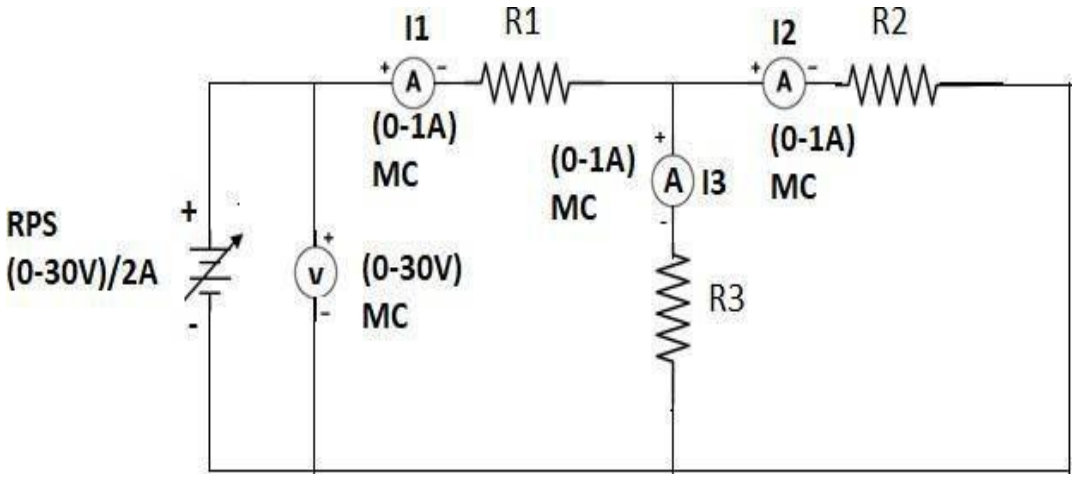
RESULT:

VIVA QUESTIONS

1. What are ohmic resistance? Give two examples.
2. What is S.I. Unit of resistance?
3. Why ammeter connected in series only?
4. Why the internal resistance of voltmeter is very high, where as that of an ammeter is very low?
5. What is the function of rheostat in the circuit?
6. What is the application of Ohm`s law?
7. What are the limitations of Ohm`s law?
8. What is the difference between emf and terminal voltage of cell?
9. State Ohm`s law.
10. What are non-ohmic resistance? Give two examples.

KIRCHHOFF'S CIRCUIT LAWS

CIRCUIT DIAGRAM FOR KCL:



TABULOUR COLUMN FOR KCL:

S.No.	Applied Voltage(V)	Theoretical			Practical		
		I_1(A)	I_2(A)	I_3(A)	I_1(A)	I_2(A)	I_3(A)

EXPERIMENT NO:2

KIRCHHOFF'S LAWS

AIM: To Verify Kirchhoff's Current Law and Kirchhoff's Voltage Law for the given circuit.

APPARATUS:

SI. NO	NAME OF THE EQUIPMENT	TYPE	RANGE	QUANTITY
1	Regulated Power Supply	-	0-30V/2A	1 NO
2	Voltmeter	MC	0-30V	4 NO
3	Ammeter	MC	0-1A	3 NO
4	Wooden Resistor Box	WW	22Ω,50Ω,75Ω	Each 1NO
5	Connecting Wires			Required

THEORY:

KCL: The algebraic sum of all the currents meeting at a node or junction is equal to zero or the sum of currents flowing into that node is equal to the sum of currents flowing out of that node.

The current entering any junction is equal to the current leaving that junction, $I_1=I_2 + I_3$

The law is based on the conservation of charge. The current law is applicable to any lumped network irrespective

of the nature of the network; whether unilateral or bilateral, active or passive, linear or non-linear.

KVL: In any closed path / mesh, the algebraic sum of EMF and voltage drops is zero. The sum of all the voltages around a loop is equal to zero i.e. V =v1 + v2 + v3

Kirchhoff's Voltage Law (KVL) is Kirchhoff's second law that deals with the conservation of Energy around a closed-circuit path.

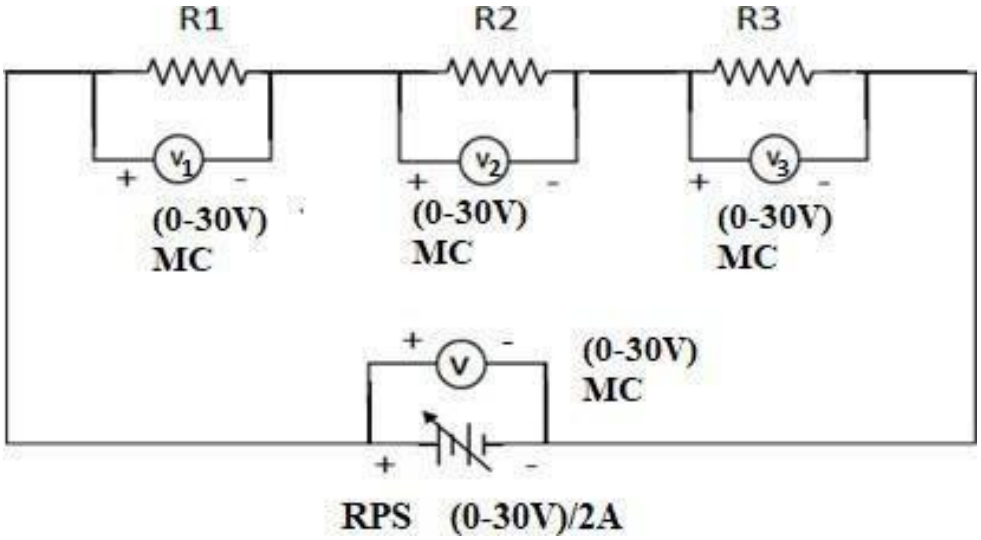
PROCEDURE:

1. Make circuit connections as per circuit diagram.
2. Switch ON RPS and vary the voltage such that V=10V, 15V.
3. Note down the corresponding meter readings and tabulate.
4. Bring back the RPS to zero position and switch OFF the RPS.
5. Remove all connections.

FORMULA FOR KCL:

$$I_1= I_2 + I_3$$

CIRCUIT DIAGRAM FOR KVL:



TABULOUR COLUMN FOR KVL:

S.No.	Applied Voltage(V)	Theoretical			Practical		
		V_1(V)	V_2(V)	V_3(V)	V_1(V)	V_2(V)	V_3(V)

PROCEDURE:

1. Make circuit connections as per circuit diagram.
2. Switch ON RPS and vary the voltage such that V = 10V, 20V.
3. Note down the corresponding meter readings and tabulate.
4. Bring back the RPS to zero position and switch OFF the RPS.
5. Remove all connections.

FORMULA FOR KVL:

$$V = V_1 + V_2 + V_3$$

THEORITICAL CALCULATIONS:

PRECAUTIONS:

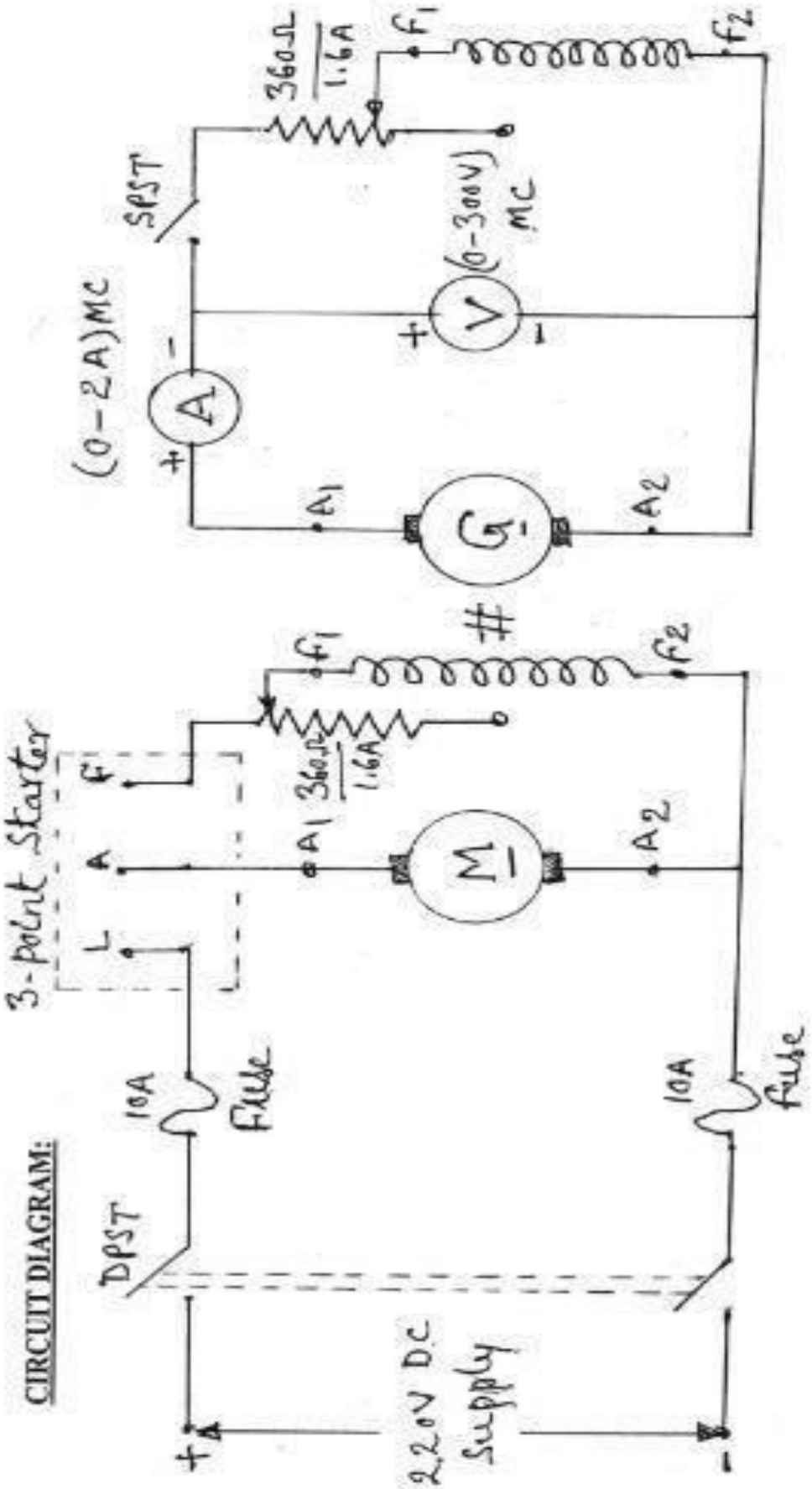
1. Connections should be very tight.
2. Readings should be taken without any parallax error.
3. Meters should be placed in horizontal positions.
4. Avoid short circuits.

RESULT:

VIVA QUESTIONS

1. What is another name for KCL & KVL?
2. Define network and circuit?
3. What is the Property of Inductor & Capacitor?

MAGNETIZATION CHARACTERISTICS OF DC SHUNT GENERATOR


CIRCUIT DIAGRAM:
3-Point Starter
DPST
SPST
10A Fuse
10A Fuse
360Ω / 1.6A
360Ω / 1.6A
(0-2A) MC
(0-300V) MC
220V DC Supply
A1
A2
F1
F2
L
A
F
G1
M1
V
A
+
-

EXPERIMENT NO: 3

MAGNETIZATION CHARACTERISTICS OF DC SHUNT GENERATOR

AIM: To draw the magnetization curves of a given DC shunt generator.

APPARATUS:

S.No	Name of Apparatus	Type	Range	Qty
1.	Rheostat	WW	360Ω /1.6A	2 No
2.	Voltmeter	MC	(0-300) V	1 No
3.	Ammeter	MC	(0-2) A	1 No
4.	Tachometer	Digital	0 -9999 rpm	1 No
5.	S.P.S.T	-	-	1 No

THEORY:

Open Circuit characteristics curve is the relation between field current and the no load induced E.M.F. It resembles the B-H curve of a magnetic material. Due to residual magnetism in the poles some E.M.F is generated even when I_f is zero. Hence the curve starts a little way up. The slight curvature at the lower ends is due to the magnetic inertia. The first part of the curvature is practically straight line. This is due to the fact that at low flux densities reluctance of the iron path begin negligible. The generated E.M.F is directly proportional to the field current. At higher flux densities where I is small iron path reluctance becomes appreciable and relation between E and I, no longer holds good. After point B saturation of poles starts.

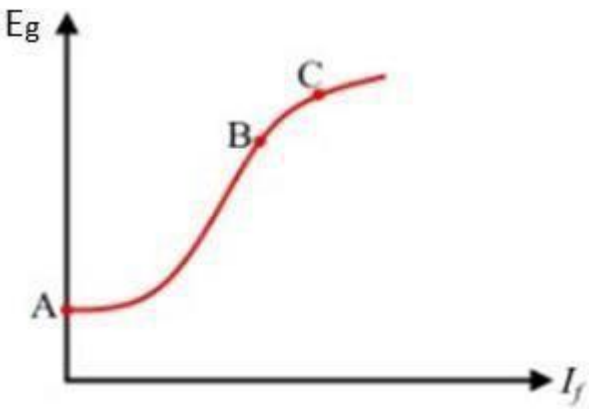
Critical field resistance is another important point to be considered to build up voltage in the generator. The value of the resistance represented by the tangent to the curve is known as “Critical field resistance” Rc for a given speed.

TABULAR COLUMN:

S.No	Rated speed (1500rpm)	
	Reading of Ammeter I_f (amps)	Reading of Voltmeter E_g (volts)
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

GRAPH:

Draw O.C.C taking field current I_f on X-axis and induced EMF (E) on Y-axis.



PROCEDURE:

1. Connect the circuit as shown in the circuit diagram.
2. Start the motor with the help of 3- point starter and then by using field rheostat make the motor to rotate at rated speed.
3. Note down the readings of the voltmeter by keeping the switch 'S' in open position
4. The voltmeter gives deflection due to presence of residual flux.
5. Close the switch 'S', gradually decrease the resistance in the field Circuit of the generator
6. Increase the field current 'I_f' so that the induced emf also increases.
7. Vary the rheostat in steps until the generator generates rated voltage.
8. At each step of field current 'If' note voltmeter reading.
9. Repeat the same procedure for different steps.

PRECAUTIONS:

1. Connections should be made tightly.
2. Observe the readings without having any parallax error.
3. Avoid short circuits.
4. Place the voltmeter and Ammeter in sleeping position.

RESULT:

VIVA QUESTIONS

1 What is residual magnetism?

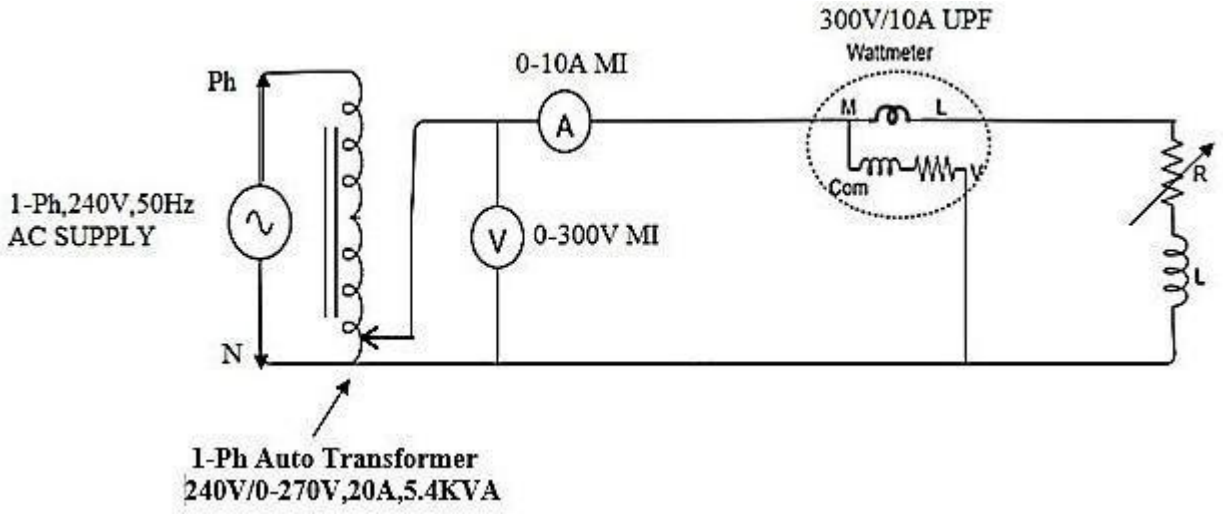
2 What is critical resistance?

3 What will happen if the field circuit is opened when motor is running?

4 Conditions for building up of voltage?

MEASUREMENT OF POWER AND POWER FACTOR

CIRCUIT DIAGRAM:



TABULAR FORM:

Sl. No.	V (in volts.)	I (in Amp.)	P (in watts.)	$\mathbf{Cos\phi = \frac{P}{VI}}$

EXPERIMENT NO: 4

MEASUREMENT OF POWER AND POWER FACTOR

AIM: Measurement of power and power factor in a Single-phase AC Series R-L circuit.

APPARATUS:

Sl. No.	Name of the Apparatus	Range	Type	Quantity
1	Single-phase Auto transformer	5.4 KVA, 240V/(0270)V,20A	Variable	01
2	Voltmeter	(0-300)V	MI	01
3	Watt-meter	300V/10A UPF	DM	01
4	Ammeter	(0-10)A	MI	01
5	Resistive Load	1-Ph,240V,10A	Variable	01
6	Inductor	1-Ph Transformer HV Winding		01
7	Connecting Wires	PVC Insulated Copper		As per required

THEORY:

In R-L series AC circuit a resistor of resistance R ohm, and Inductor of inductance L henry are connected across single phase ac supply of V volts as shown in above fig

Power consumed by the Inductive load is given by

$P=VI\cos\phi$, power factor $(\cos\phi) = P/VI$

Where, P is active power in watt, V is supplied voltage in volts, I is current flowing through the circuit elements in Amp.

Thus, by using above formula the power consumed by the load and power factor can be determined.

PROCEDURE:

1. Connect circuit as per circuit diagram.
2. Before switch on the main power supply make sure that single-phase auto transformer knob is at zero position.
3. Switch ON the power supply.
4. Now slowly increase the supply voltage to the circuit after giving supply to the single-phase Auto- transformer.
5. Take all the corresponding readings of the connected instruments in the circuit as per Observation table.
6. Now find the power factor (Cosϕ) and as per formula given in observation table.

PRECAUTIONS:

1. Don't switch ON power supply without concerning teacher.
2. Single phase Autotransformer must be kept at minimum potential point before switch ON the experiment.
3. Readings should be taken without parallax error.

SAMPLE CALCULATION:

Applied Voltage (V) = _____

Load Current (I) = _____

Power (P) = _____

power factor (Cosϕ) =P/VI

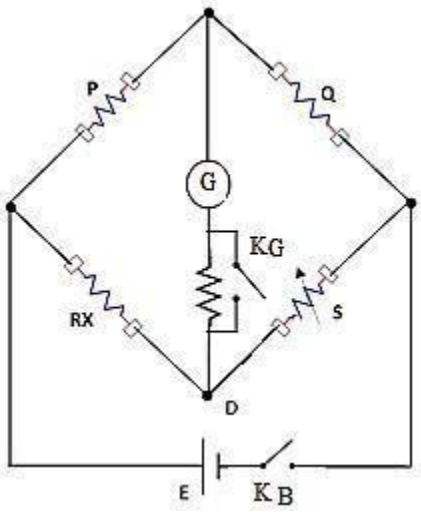
RESULT:

VIVA QUESTIONS

1. What is the measurement of power and power factor?
2. What is the unit of power factor?
3. What is the basic measurement of power?
4. What is method of measuring power?

WHEATSTONE BRIDGE

CIRCUIT DIAGRAM:



FRONT PANEL DIAGRAM:



EXPERIMENT NO: 5

WHEATSTONE BRIDGE

AIM: To measure the given medium resistance using Wheatstone bridge.

APPARATUS:

S.No.	Name of The Equipment	Quantity
1	Wheat stone Bridge kit	1 NO
2	Unknown resistance	1 NO
3	Connecting Wires	AS PER REQUIRED

THEORY:

A Wheatstone bridge is an electrical circuit used to measure an unknown electrical resistance by balancing two legs of a bridge circuit, one leg of which includes the unknown component. The primary benefit of the circuit is its ability to provide extremely accurate measurements (in contrast with something like a simple voltage divider). Its operation is similar to the original potentiometer. The Wheatstone bridge was invented by Samuel Hunter Christie (sometimes spelled "Christy") in 1833 and improved and popularized by Sir Charles Wheatstone in 1843. One of the Wheatstone bridge's initial uses was for soil analysis and comparison.

PROCEDURE:

1. Make the connections of coil is connected between X1 and X2.
2. The P/Q ratio (multiplier) is initially kept at position '1' and the deflection of the galvanometer is observed by pressing both the battery and the galvanometer keys.
3. The S arm (X 1000Ω) is adjusted and two positions are identified for which the deflection of the galvanometer is on either side of the null point and kept at the lowest value of S. Then the x100Ω, x10Ω, x1Ω knobs of S are adjusted to get null deflection. If necessary, the sensitivity knob may be controlled to get appreciable deflection. [If not possible P/Q ratio is kept at suitable value i.e., any one of ratios provided.
4. The value of unknown resistance is read. (S value)
5. Steps 3 and 4 are repeated for some other P/Q ratio. The mean value is taken.
6. The experiment is repeated with other samples provided.

TABULAR FORM:

S.No.	P/Q Ratio (Multiplier)	S Value (Ω)	Unknown Resistance Rx = P/Q * S (Ω)	Average resistance of coil(Ω)

SAMPLE CALCULATION:

Unknown Resistance, Rx = P/Q * S (Ω)

Where P/Q = Ratio Arms

S = Variable resistance

Rx = Unknown resistance

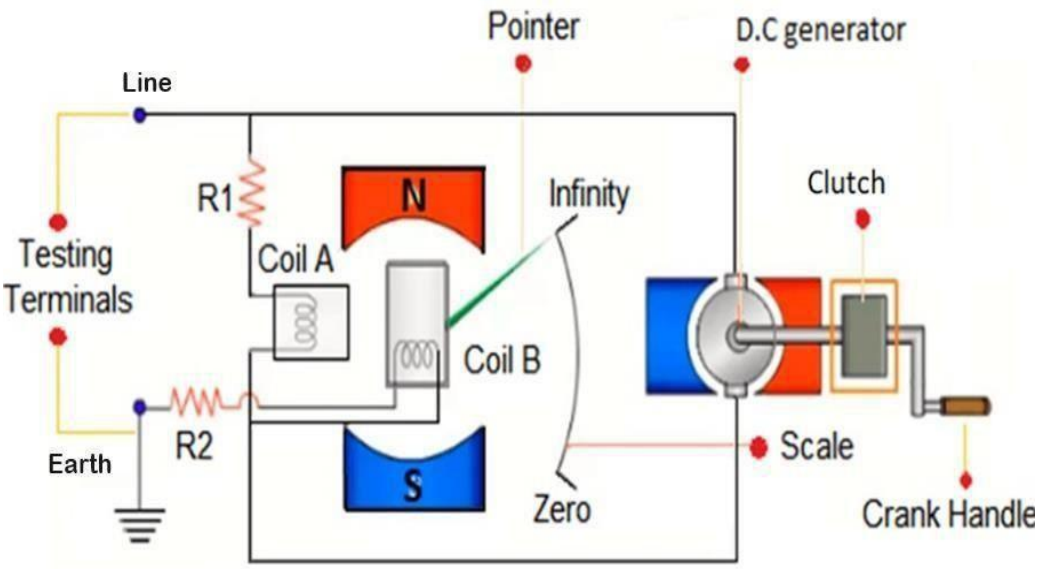
RESULT:

VIVA QUESTIONS

1. Which law is used in Wheatstone bridge?
2. What is the full name of the Wheatstone bridge?
3. Which stone bridge formula?
4. Why is Wheatstone bridge so called?

MEGGER

CIRCUIT DIAGRAM:



TABULAR FORM FOR WINDING TO EARTH:

S.No	3-Ph Induction Motor Stator winding terminals	Insulation Resistance(MΩ)
1	U_1 to Earth	
2	V_1 to Earth	
3	$W_1$to Earth	
4	U_2 to Earth	
5	V_2 to Earth	
6	$W_2$to Earth	

TABULAR FORM FOR BETWEEN WINDINGS:

S.No	3-Ph Induction Motor Stator winding terminals	Insulation Resistance(MΩ)
1	U_1 to V1	
2	V_1 to W1	
3	$W_1$to U1	
4	U_2 to V2	
5	V_2 to W2	
6	$W_2$to U2	

EXPERIMENT NO:6

MEGGER

AIM: Measurement of Insulation Resistance using Megger.

APPARATUS:

S.No.	Name of The Apparatus	Quantity
1	Megger 500V	01
2	Connecting Wires	02

THEORY:

A Megger is used to measure insulation resistance and it is powered by a hand driven inbuilt DC generator or battery of a higher voltage range, it is called Mega ohmmeter. The device enable us to measure electrical leakage in wire, results are very reliable as we shall be passing electric current through device while we are testing. The equipment basically uses for verifying the electrical insulation level of any device such as motors, cables, generators, windings, etc. This is a very popular test being carried out since very long back. Not necessary it shows us exact area of electrical puncture but shows the amount of leakage current and level of moisture within electrical equipment/winding/system.

CONSTRUCTION:

Megger is a DC generator. It consists of three terminals i.e., Line terminal, Guard terminal, and the Earth terminal. In the above circuit, the guard is connected on top of an insulator, the line terminal is connected to the conductor, which is to be tested, and the earth pin is grounded.

Higher Resistance = Higher Insulation = No current flow.

PROCEDURE:

1. Connect the Line terminal of the Megger is connected to winding terminal of the Induction Motor and Earth terminal of the Megger is connected to earth wire.
2. The handle of the Megger is rotated at about 160 rpm by hand thus the Megger will generate about 100 V to 500 V DC.
3. The current flows through Induction Motor, the resistance in scale is noted which ranges between 35 to 100Mega Ohms.
4. Note to maintain this contact for 30 to 60 seconds.

PRECAUTIONS:

1. Don’t touch the line terminal of the Megger at the time of rotating Handle.
2. Reading Should be taken carefully.

RESULT:

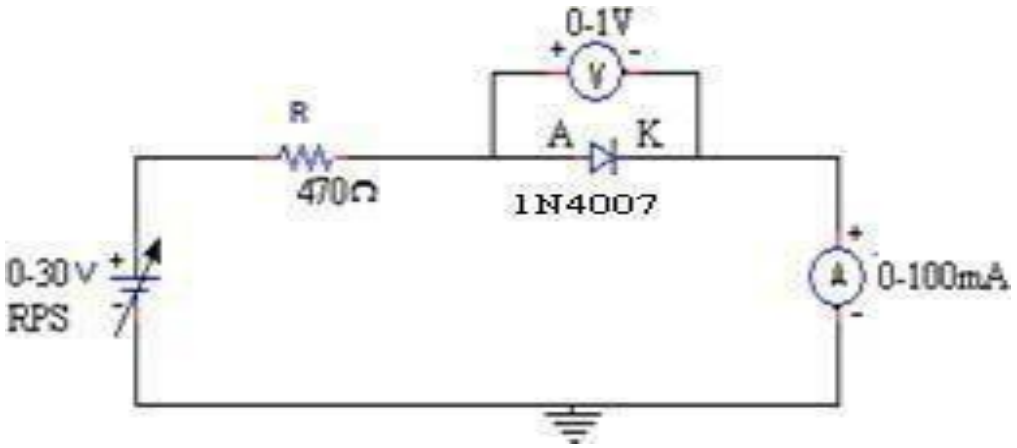
VIVA QUESTIONS

1. What is the working principle of Megger?
2. Why is Megger used?
3. What is the limit of Megger?
4. How to make a Megger test report?

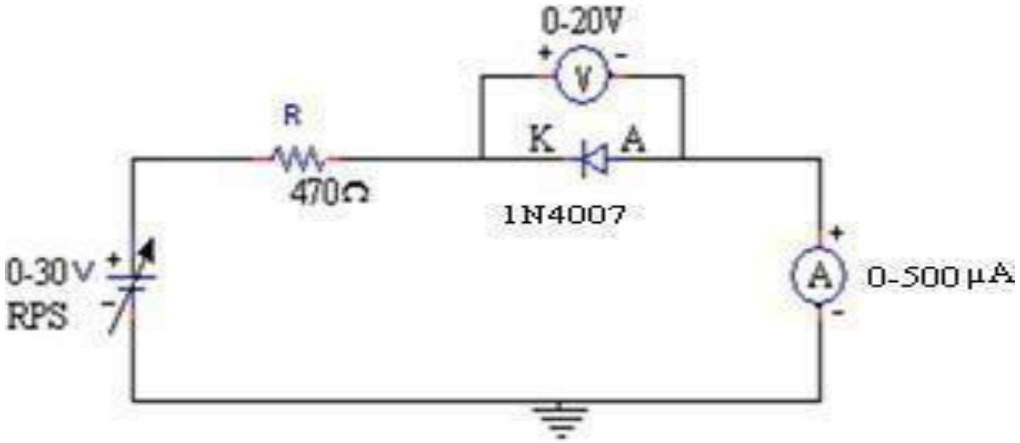
V-I CHARACTERISTICS OF PN JUNCTION DIODE

CIRCUIT DIAGRAM: SILICON:

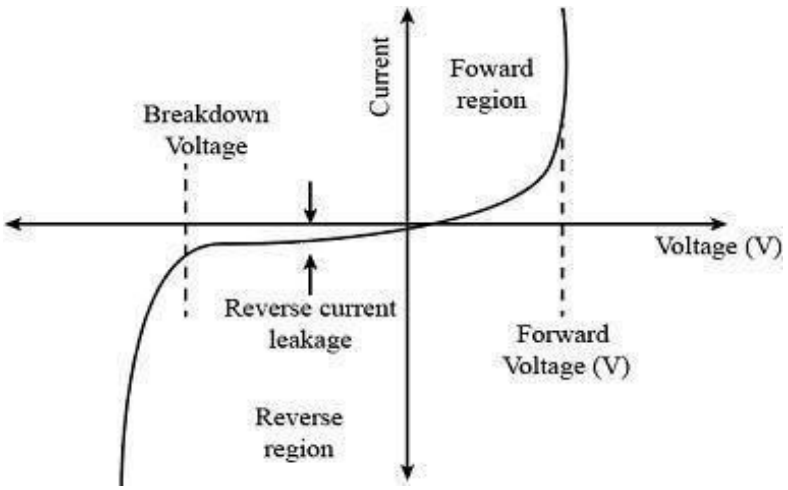
FORWARD BIAS:



REVERSE BIAS:



MODEL GRAPH:



EXPERIMENT NO.7

V-I CHARACTERISTICS OF PN JUNCTION DIODE

AIM: To observe the voltage-current relationship in a PN junction diode and to calculate the static and dynamic resistances of the diode.

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	PN junction diode(Si)	1N4007	1
2.	Resistor	470Ω	1
3.	Bread board	General purpose	1
4.	DC voltmeter	0-1V	1
		0-20V	1
5.	DC ammeters	0-100mA	1
		0-500μA	1
6.	Regulated power supply	0-30V(DC)	1
7.	Connecting wires	1set	few

THEORY:

A p-n junction diode conducts only in one direction. The V-I characteristics of the diode are curve between voltage across the diode and current through the diode. When external voltage is zero, circuit is open and the potential barrier does not allow the current to flow. Therefore, the circuit current is zero. When P-type (Anode)is connected to +ve terminal and n- type (cathode) is connected to –ve terminal of the supply voltage is known as forward bias. The potential barrier is reduced when diode is in the forward biased condition. At some forward voltage, the potential barrier altogether eliminated and current starts flowing through the diode and also in the circuit. The diode is said to be in ON state. The current increases with increasing forward voltage. When N- type (cathode) is connected to +ve terminal and P-type (Anode) is connected –ve terminal of the supply voltage is known as reverse bias and the potential barrier across the junction increases. Therefore, the junction resistance becomes very high and a very small current (reverse saturation current) flows in the circuit. The diode is said to be in OFF state. The reverse bias current due to the minority charge carriers.

SILICON DIODE TABULAR FORMS:FORWARD BIAS

S.No	Forward bias voltage(Vf) in volts	Forward bias current(If) in mA
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

REVERSE BIAS

S.No	Reverse bias voltage(Vf) in volts	Reverse bias current(If) in μA
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Cut-in voltage

By the term cut-in voltage, indicated by V , is meant the voltage below which the current is very small.As the voltage exceeds cut-in voltage, the current increases very rapidly. The cut-in voltage is sometimes called offset voltage, break point voltage or threshold voltage.

Cut-in voltage for germanium diode is 0.2 V and about 0.6V for silicon diode.

Dynamic (ac) resistance

The a.c. resistance of a diode, at a particular D.C. voltage, is equal to the reciprocal of the slope of the characteristic at that point, i.e. the a.c. resistance, $r_{ac} = \Delta V/\Delta I$.

The resistance offered by the diode to an a.c. signal is called its dynamic or a.c. resistance. The resistance we calculated using the relation $\Delta V/\Delta I$ is called the ac or dynamic resistance(r).

Hence we will refer to the ac resistance of the diode as r_D where the lower case letter r is in keeping with the

convention of using lower case letters for ac quantities. Thus $r_{ac}=\Delta V/\Delta I$.

The a.c. resistance of a diode, at a particular D.C. voltage, is equal to the reciprocal of the slope of the characteristic at that point. The value of a.c. resistance of a diode is in range of 1 to 25Ω.usually, it is smaller than the D.C. Resistance of a diode.

Static (dc) resistance

When dc voltage is applied across the diode certain dc current will flow through it. The dc resistance of a diode is found by dividing the dc voltage across it by the dc current through it. Thus the dc resistance also called the static resistance is found by direct application of ohms law. This denoted by R_D .$R_D = V/I$

The static resistance of diode is quite low (in ohms)

APPLICATIONS:

Diodes are used in

1. Clamping circuits for DC restoration.
2. Clipping circuits for wave shaping.
3. Voltage multipliers.
4. Rectifiers.

PROCEDURE:

Forward Characteristics:

1. Connect the circuit as per the circuit diagram.
2. Increase source voltage V_F in steps of 0.1 V.
3. Note down the corresponding Ammeter readings for every incremented value of V_F.
4. Tabulate the values and plot a graph with V_F on x-axis and I_F on y-axis.

Reverse Characteristics:

1. Connect the circuit as per the circuit diagram.
2. Increase source voltage V_R in steps of 0.5V.
3. Note down the corresponding Ammeter readings for every incremented value of V_R.

CALCULATIONS:

Calculation of Static and Dynamic Resistance for a given diode.

In forward bias condition:

Static Resistance , $R_S = V_f/I_f =$

Dynamic Resistance, $R_D = \Delta V_f/\Delta I_f =$

In Reverse bias condition:

Static Resistance , $R_S = V_R/I_R =$

DynamicResistance, $R_D = \Delta V_R/\Delta I_R =$

PRECAUTIONS:

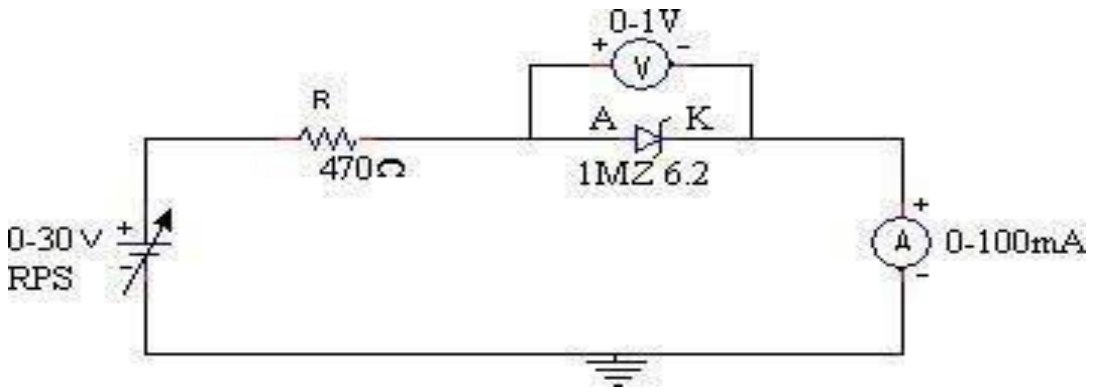
1. Avoid loose connections.
2. Care must be taken that all the sources and meters are connected with correct polarity.
3. Before connecting the meter, ensure that it indicates “zero” reading.
4. Voltmeter reading should not exceed 0.9v for forward bias and 15v for reverse bias.

RESULT:

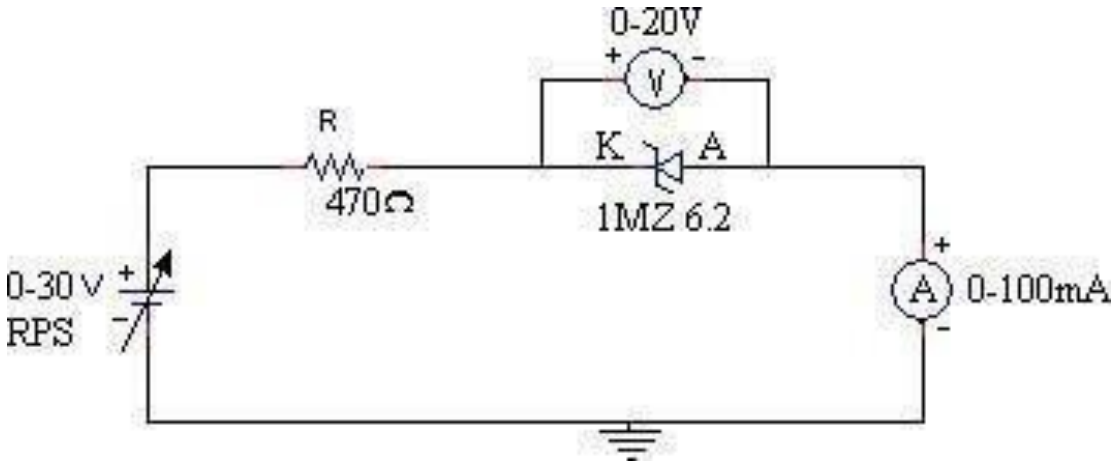
V-I CHARACTERISTICS & VOLTAGE REGULATOR OF ZENER DIODE

CIRCUIT DIAGRAM:

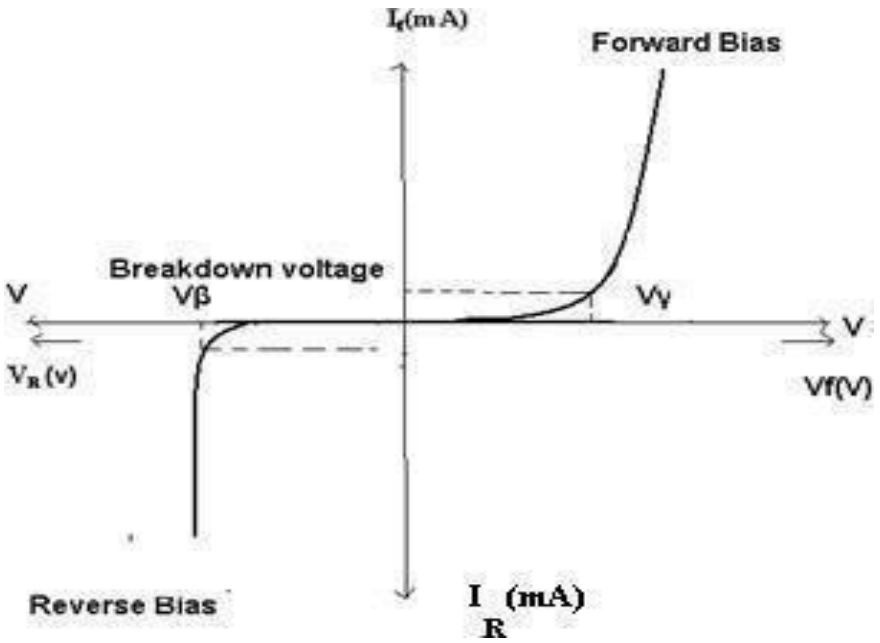
FORWARD BIAS



REVERSE BIAS



MODEL GRAPH:



EXPERIMENT NO: 8

V-I CHARACTERISTICS &VOLTAGE REGULATOR OF ZENER DIODE

AIM: a) To observe and draw the static characteristics of a zener diode.
. b)To find the voltage regulation of a given zener diode.

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Zener diode	1MZ6.2V, 3watts	1
2.	Resistor	470Ω 1KΩ	1 1
3.	Decade resistance box		1
4.	Bread board	General purpose	1
5.	DC voltmeter	0-1V	1
		0-20V	1
6.	DC ammeters	0-100mA	2
7.	Regulated power supply	0-30V (DC)	1
8.	Connecting wires	1 set	Few

THEORY:

A zener diode is heavily doped p-n junction diode, specially made to operate in the break down region. A p-n junction diode normally does not conduct when reverse biased. But if the reverse bias is increased, at a particular voltage it starts conducting heavily. This voltage is called Break down Voltage. High current through the diode can permanently damage the device.

To avoid high current, we connect a resistor in series with zener diode. Once the diode starts conducting it maintains almost constant voltage across the terminals whatever may be the current through it, i.e., it has very low dynamic resistance. It is used in voltage regulators.

TABULAR FORMS: FORWARD BIAS

S.No	Forward bias voltage(Vf) in volts	Forward bias current(If) in mA
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

REVERSE BIAS

S.No	Reverse bias voltage(Vf) in volts	Reverse bias current(If) in mA
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

APPLICATIONS:

1. Voltage regulation.
2. Reference elements.
3. Surge suppressors.
4. Switching applications and clipper circuits.

PROCEDURE:

Forward characteristics

1. Connect the circuit as per the circuit diagram.
2. Increase source voltage V_S in steps of 0.1V.
3. Note down the corresponding Ammeter and Voltmeter readings for every incrementedvalue of V_F.
4. Tabulate the values and plot a graph with V_F on x-axis and I_F on y-axis.

Reverse characteristics

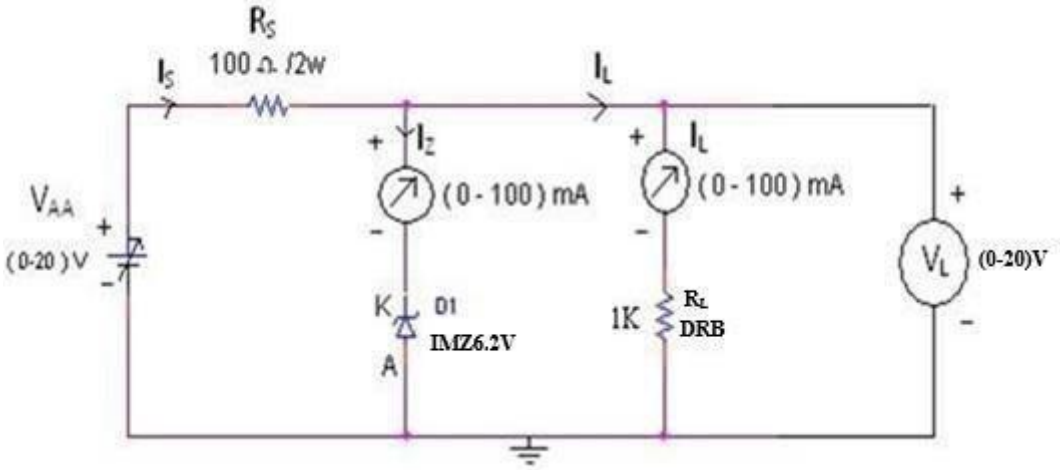
1. Connect the circuit as per the circuit diagram.
2. Increase source voltage V_S in steps of 0.5 V
3. Note down the corresponding Ammeter and voltmeter readings for every increment of V_S.
4. Tabulate the values and plot a graph with V_R on x-axis and I_R on y-axis.

PRECAUTIONS:

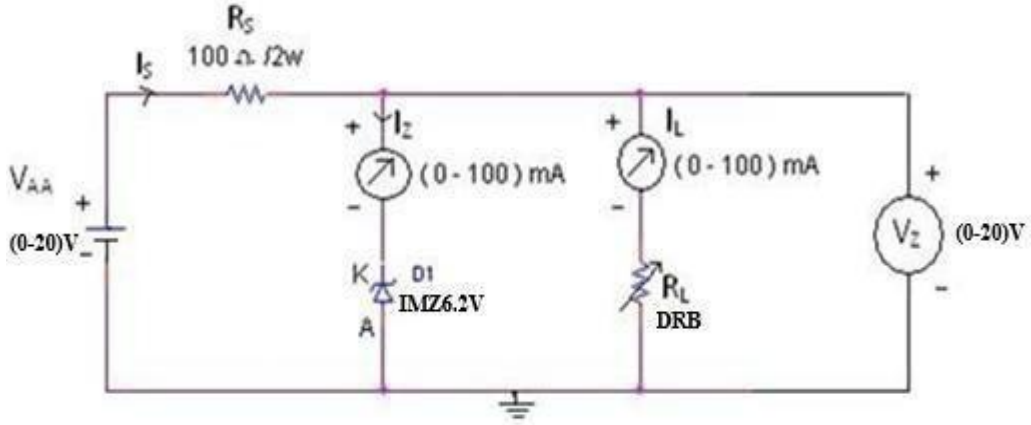
1. Avoid loose connections.
2. Care must be taken that all the sources and meters are connected with correct polarity.
3. Before connecting the meter, ensure that it indicates "zero" reading.

CIRCUIT DIAGRAM

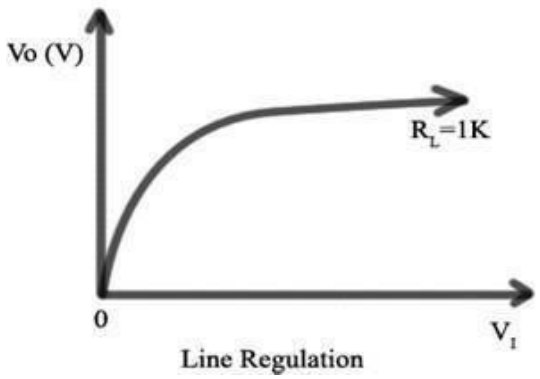
LINE REGULATION



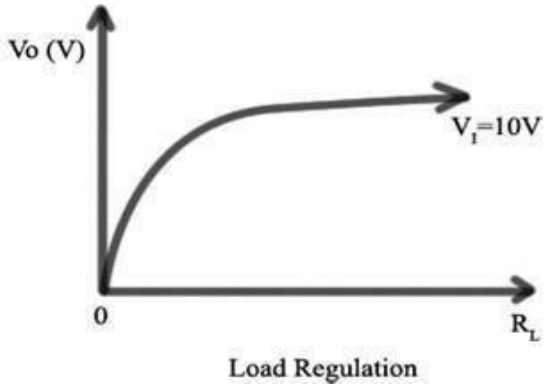
LOAD REGULATION



MODEL GRAPH:



Line Regulation



Load Regulation

ZENER DIODE AS A VOLTAGE REGULATOR

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Zener diode	1MZ6.2V, 3watts	1
2.	Resistor	100Ω	1
3.	Decade resistance box		1
4.	Bread board		1
5.	DC voltmeter	0-1V	1
		0-20V	1
6.	DC ammeters	0-100mA	2
7.	Regulated power supply	0-30V(DC)	1
8.	Connecting wires		Few

THEORY:

Zener diode is a P-N junction diode specially designed to operate in the reverse biasedmode. It is acting as normal diode while forward biasing. It has a particular voltage known as break down voltage, at which the diode break downs while reverse biased. In the case of normal diodes the diode damages at the break down voltage. But Zener diode is specially designed to operate in the reverse breakdown region.The basic principle of Zener diode is the Zener breakdown. When a diode is heavily doped, it's depletion region will be narrow. Whena high reverse voltage is applied across the junction, there will be very strong electric field at the junction. And the electron hole pair generation takes place. Thus heavy current flows.This is known as Zener break down. So a Zener diode, in a forward biased condition acts as a normal diode. In reverse biased mode, after the break down of junction current through diode increases sharply. But the voltage across it remains constant. This principle is used in voltage regulator using Zener diodes. The figureshows the zener voltage regulator, it consists of a current limiting resistor RS connected in series with the input voltage Vs and zener diode is connected in parallel with the load RL in reverse biased condition. The output voltage is always selected with a breakdown voltage Vz of the diode. The input source current,

$I_S = I_Z + I_L$.....(1)

TABULAR FORMS:

LINE REGULATION

S.No	Input voltage Vi (V)	Zener current IZ (mA)	Load current IL (mA)	Output voltage VO(V)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				

LOAD REGULATION

S.No	Load resistance RL (V)	Zener current IZ (mA)	Load current IL (mA)	Output voltage VO(V)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

The drop across the series resistance, Rs = Vin – Vz ..... (2

And current flowing through it, Is = (Vin – VZ) / RS .....(3)

Regulation with a varying input voltage (line regulation)

It is defined as the change in regulated voltage with respect to variation in line voltage. It is denoted by 'LR'.In this, input voltage varies but load resistance remains constant hence, the load current remains constant. As the input voltage increases, form equation (3) Is also varies accordingly. Therefore, zener current Iz will increase. The extra voltage is dropped across the Rs. Since, increased Iz will still have a constant Vz and Vz is equal to Vout. The output voltage will remain constant. If there is decrease in Vin, Iz decreases as load current remains constant and voltage drop across Rs is reduced. But even though Iz may change,Vz remains constant hence, output voltage remains constant.

Regulation with the varying load (load regulation)

It is defined as change in load voltage with respect to variations in load current. To calculate this regulation, input voltage is constant and output voltage varies due to change in the load resistance value. Consider output voltageis increased due to increasing in the load current. The left side of the equation (4) is constant as input voltage Vin, IS and Rs is constant. Then as load current changes, the zener current Iz will also change but in opposite way such that the sum of Iz and IL will remain constant. Thus, the load current increases, the zener current decreases and sum remain constant. Form reverse bias characteristics even Iz changes, Vz remains same hence, and output voltage remains fairly constant.

PROCEDURE:-

A) Line Regulation

1. Make the connections as shown in figure below.
2. Keep load resistance value at 1K, vary DC input voltage from 5V to 15V.
3. Note down output voltage as a load voltage with high line voltage 'V_{HL}' and as a loadvoltage with low line voltage 'V_{LL}'.
4. Using formula, % Line Regulation = (V_{HL}-V_{LL})/ V_{NOM} x100, where V_{NOM} = the nominal loadvoltage under the typical operating conditions. For ex. V_{NOM} =
5. Line Regulation = (VHL-VLL)/VNOM*100 =

B) Load Regulation

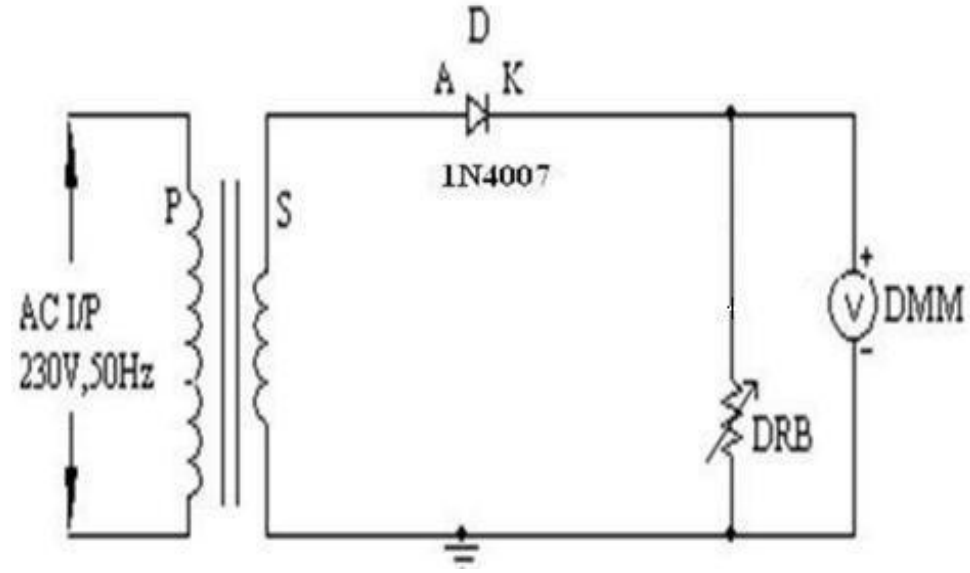
1. For finding load regulation, make connections as shown in figure below.
2. Keep input voltage constant say 10V, vary load resistance value from 100Ω-500Ω and 1K-10k.
3. Note down no load voltage 'V_{NL}' for maximum load resistance value and full load voltage'V_{FL}' for minimum load resistance value.
4. Calculate load regulation using, % load regulation = $(V_{NL}-V_{FL})/ V_{FL}$ x100
5. Load regulation = VNL -VFL / VFL*100=

RESULT:

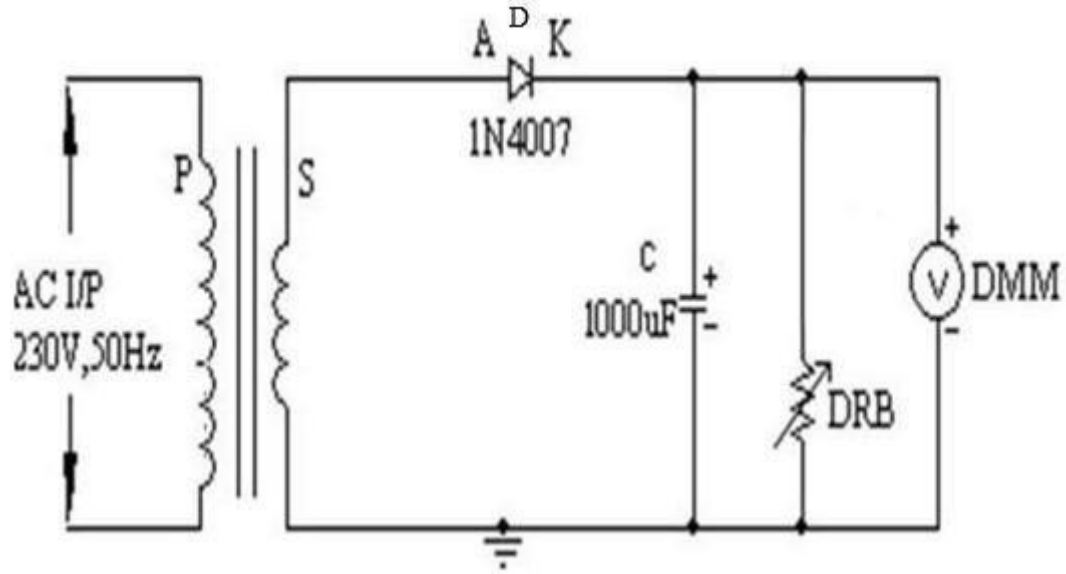
HALF WAVE RECTIFIER

CIRCUIT DIAGRAM:

A) Half Wave Rectifier Without Filter:



B) Half Wave Rectifier With Filter



EXPERIMENT NO: 9

HALF WAVE RECTIFIER

AIM: To examine the input and output waveforms of half wave Rectifier and also calculate its load regulation and ripple factor.

1. With Filter
2. Without Filter

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Step down transformer	230/6V	1
2.	Diode	1N4007	1
3.	Decade Resistance Box		1
4.	Multimeter		1
5.	Capacitor	1000µF	1
6.	Bread Board		1
7.	Connecting wires		Few

THEORY:

In half wave rectification, when ac supply is applied at the input, only positive half cycle appears across the load whereas, the negative half cycle is suppressed. How this can be explained as follows: During positive half- cycle of the input voltage, the diode d1 is in forward bias and conducts through the load resistor rl. Hence the current produces an output voltage across the load resistor rl, which has the same shape as the +ve half cycle of theinput voltage. During the negative half-cycle of the input voltage, the diode is reverse biased and there is no current through the circuit. I.e., the voltage across rl is zero. The net result is that only the +ve half cycle of the input voltage appears across the load. The average value of the half wave rectified o/p voltage is thevalue measured on dcvoltmeter.

For practical circuits, transformer coupling is usually provided for two reasons. The voltage can be stepped-up or stepped-down, as needed.

The ac source is electrically isolated from the rectifier. Thus, preventing shock hazards in the secondary circuit.

The efficiency of the half wave rectifier is 40.6%

TABULAR FORMS:

Without Filter: No load voltage V_{dcNL} = ... volts

S. No.	Load Resistance in Ohms	V_{dc}	V_{ac}	Ripple Factor Γ=Vac/vdc	% of Regulation $\frac{V_{noload} - V_{fullload}}{V_{fullload}} \times 100\%$
1	100				
2	200				
3	300				
4	400				
5	500				
6	600				
7	700				
8	800				
9	900				
10	1 k				

With Filter: No load voltage V_{dcNL} =..... volts

S. No.	Load Resistance In Ohms	V_{dc}	V_{ac}	Ripple Factor Γ=Vac/vdc	% of Regulation $\frac{V_{noload} - V_{fullload}}{V_{fullload}} \times 100\%$
1	100				
2	200				
3	300				
4	400				
5	500				
6	600				
7	700				
8	800				
9	900				
10	1 k				

Ripple factor

The ratio of the rms value of the AC component to the DC component in the output is known as"Ripple Factor".

APPLICATIONS:

Half Wave Rectifier (HWR) can be used in

1. To obtain the desired level of dc voltage by using Step up or Step down Transformer.
2. Detector in an AM radio.
3. Pulse generator circuits.
4. Voltage doublers circuits.

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.
3. The decade resistance box is set in different resistance values. By using multimeter, measure the acinput voltage of the rectifier and ac and dc voltage at the output of the rectifier.
4. Ripple factor and % regulation are calculated as per the formula Ripple factor=RMS voltage of ac components at output/DC voltage

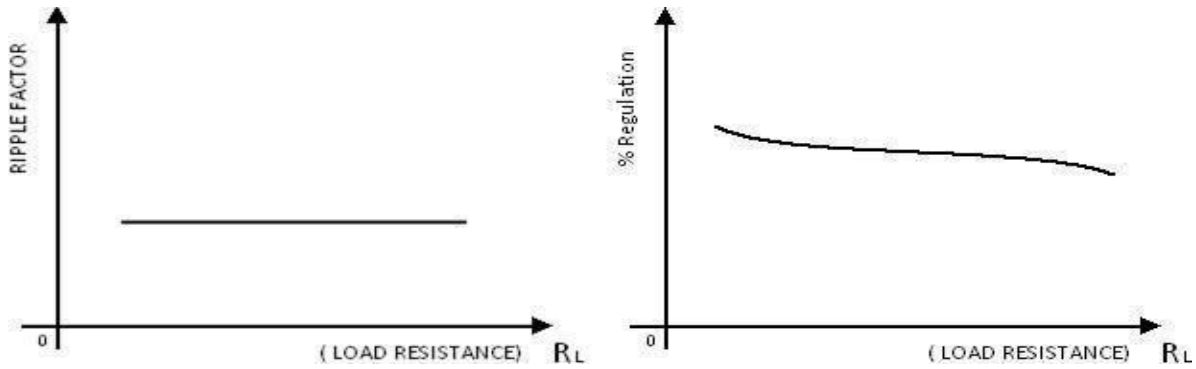
$$\%\text{Regulation} = ((V_{NL}-V_{FL})/ V_{FL}) \times 100$$

5. Capacitor filter is put in the parallel to the DRB plus 20 ohms and the above process is repeated.

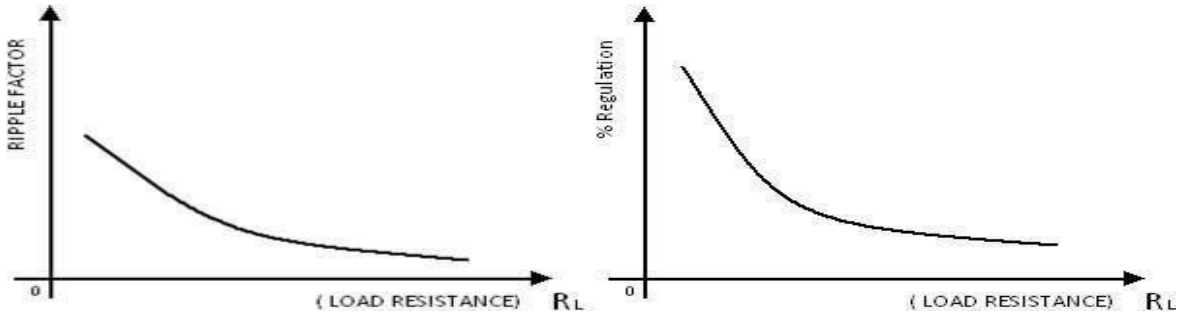
MODEL GRAPH:

LOAD CHARACTERISTICS:

With Out Filter:



With Filter



PRECAUTIONS:

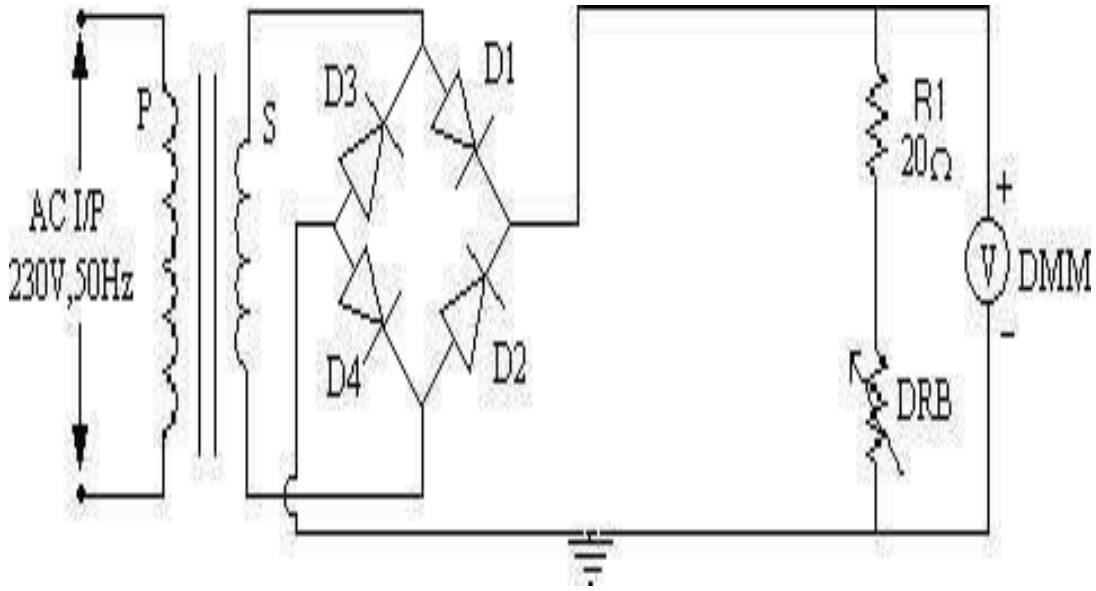
1. The primary and secondary side of the transformer should be carefully identified.
2. The polarities of all the diodes should be carefully identified.

RESULT:

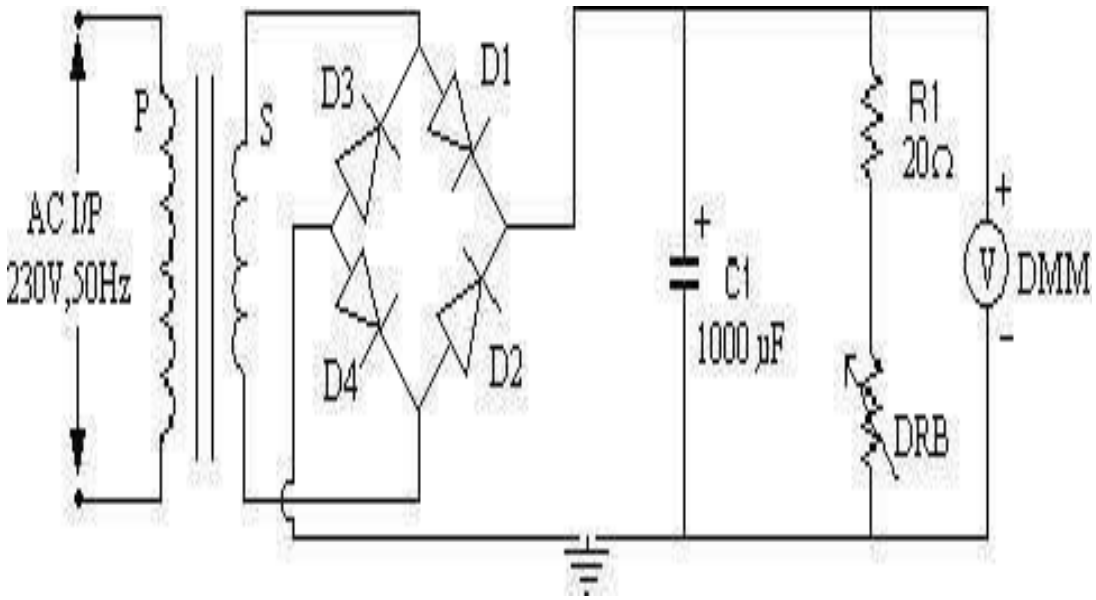
FULL WAVE RECTIFIER

CIRCUIT DIAGRAM:

A) FULL WAVE RECTIFIER WITHOUT FILTER:



B) FULL WAVE RECTIFIER WITH FILTER:



FULL WAVE RECTIFIER

AIM: To Examine the input and output waveforms of Full Wave Rectifier and also calculate its load regulation and ripple factor.

1. With Filter
2. Without Filter

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Step down transformer	230/6V	1
2.	Diode	1N4007	4
3.	Decade Resistance Box		1
4.	Multimeter		1
5.	Capacitor	1000μF	1
6.	Connecting wires		
7.	Bread Board		1

THEORY:

The circuit of a center-tapped full wave rectifier uses two diodes D1&D2. During positive half cycle of secondary voltage (input voltage), the diode D1 is forward biased and D2 is reverse biased. So the diode D1 conducts and current flows through load resistor R_L.

During negative half cycle, diode D2 becomes forward biased and D1 reverse biased. Now, D2 conducts and current flows through the load resistor R_L in the same direction. There is a continuous current flow through the load resistor R_L, during both the half cycles and will get unidirectional current as show in the model graph. The difference between full wave and half wave rectification is that a full wave rectifier allows unidirectional (one way) current to the load during the entire 360 degrees of the input signal and half-wave rectifier allows this only during one half cycle (180 degree).

TABULAR FORM:

Without Filter: No load voltage V_{dcNL} = …… volts

S. No.	Load Resistance in Ohms	V_{dc}	V_{ac}	Ripple Factor Γ=Vac/vdc	% of Regulation $= \frac{V_{noload} - V_{fullload}}{V_{fullload}} \times 100\%$
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

With Filter: No load voltage V_{dcNL} = ……volts

S. No.	Load Resistance in Ohms	V_{dc}	V_{ac}	Ripple Factor Γ=Vac/vdc	% of Regulation $= \frac{V_{noload} - V_{fullload}}{V_{fullload}} \times 100\%$
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Ripple factor

The ratio of the rms value of the AC component to the DC component in the output is known as"Ripple Factor".

APPLICATIONS:

Full Wave Rectifier (FWR) can be used in

1. DC regulated Power supplies.
2. Detect the amplitude of modulated radio signal.
3. To supply polarized voltage in welding.

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.
3. The decade resistance box is set in different resistance values. By using multimeter, measure the ac input voltage of the rectifier and ac and dc voltage at the output of the rectifier.
4. Measure DC No Load voltage by removing load resistance. Ripple factor and % regulation are calculated as per the formula:

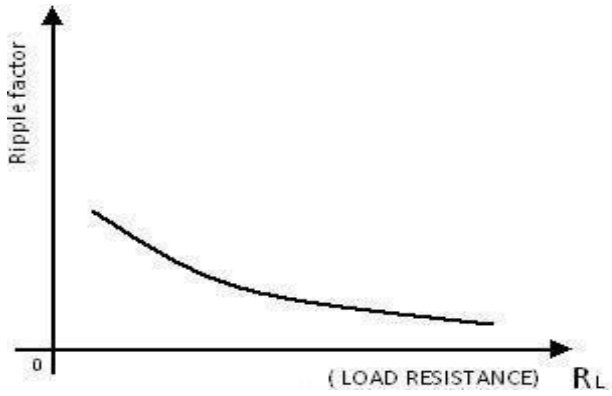
   Ripple factor=RMS voltage of ac components at output/DC voltage

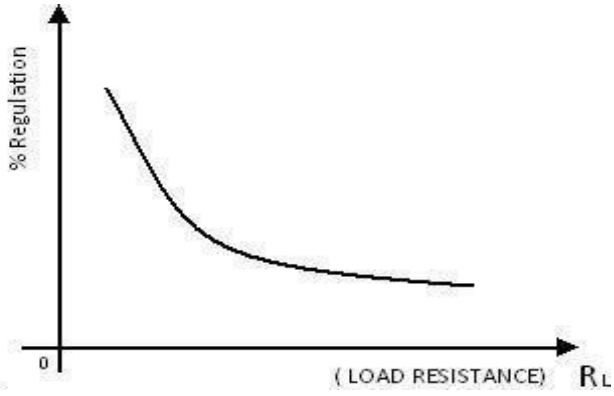
   $$\%\text{Regulation} = ((V_{NL}-V_{FL})/V_{FL})X100$$

5. Capacitor filter is put in the parallel to the DRB plus 20 ohms and process is Repeated

MODEL GRAPH:

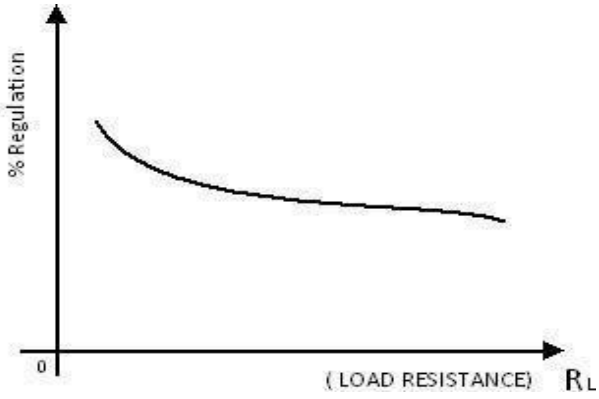
With filter





Without filter





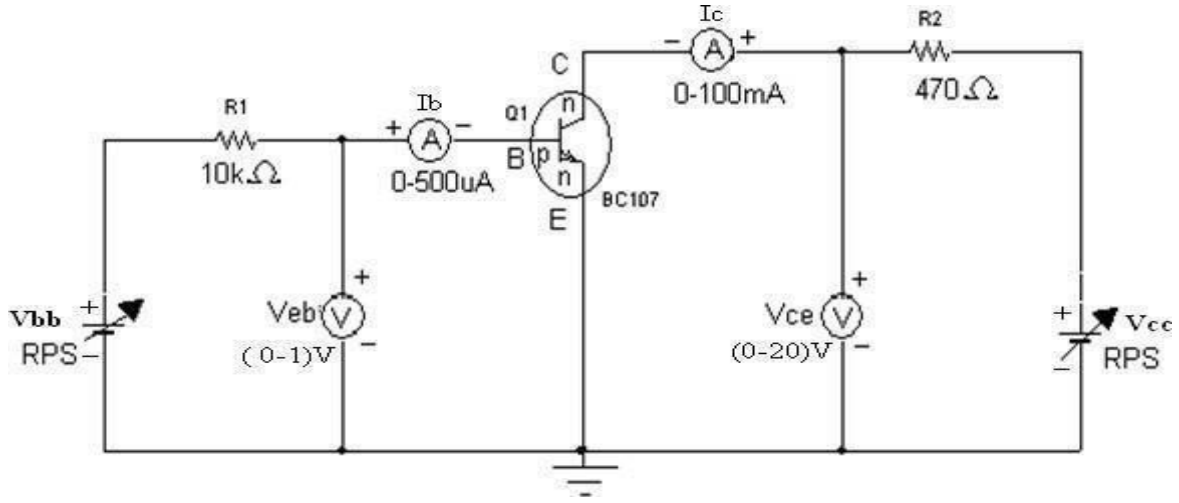
PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified.
2. The polarities of all the diodes should be carefully identified.

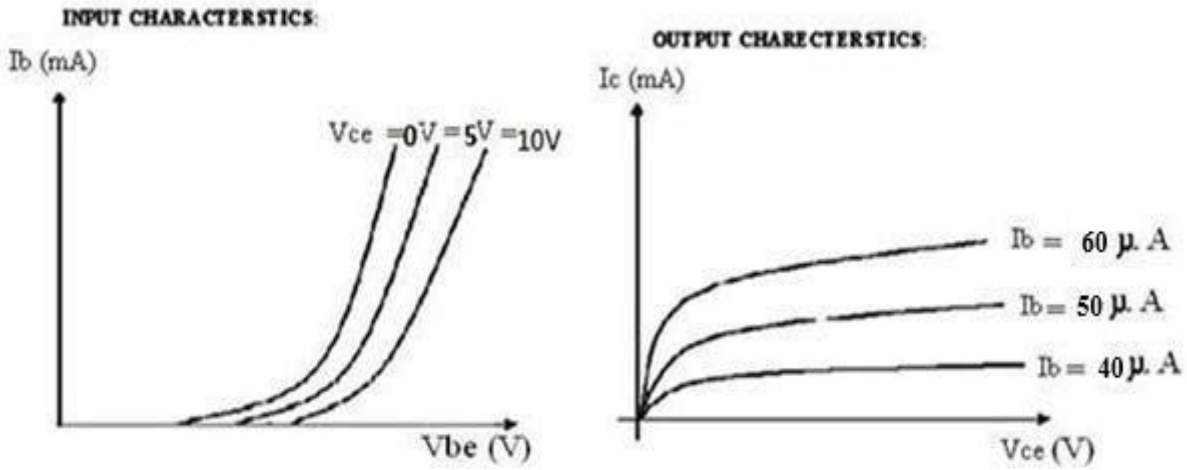
RESULT:

INPUT & OUTPUT CHARACTERISTICS OF BJT IN CE & CB CONFIGURATION

CIRCUIT DIAGRAM:



MODEL GRAPHS:



EXPERIMENT NO: 10

INPUT & OUTPUT CHARACTERISTICS OF BJT IN CE CONFIGURATION

AIM: To plot the Input and Output characteristics of a transistor connected in Common Emitter Configuration.

APPARATUS REQUIRED:

S.No	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Transistor	BC107	1
2.	Resistor	10KΩ	1
3.	Resistor	470Ω	1
4.	Bread board	General purpose	1
5.	DC voltmeter	0-1V	1
		0-20V	1
6.	DC ammeters	0-100mA	1
		0-500μA	1
7.	Regulated power supply	0-30V	2
8.	Connecting wires	1 set	Few

THEORY:

A transistor is a three terminal device. The terminals are emitter, base and collector. In common emitter configuration, input voltage is applied between base and emitter terminals and output is taken across the collector and emitter terminals. Therefore the emitter terminal is common to both input and output. The input characteristics resemble that of a forward biased diode curve. This is expected since the Base-Emitter junction of the transistor is forward biased. As compared to CB arrangement I_B increases less rapidly with V_{BE}. Therefore input resistance of CE circuit is higher than that of CB circuit.

The output characteristics are drawn between I_c and V_{CE} at constant I_B. the collector current varies with V_{CE} up to few volts only. After this the collector current becomes almost constant, and independent of V_{CE}. The value of V_{CE} up to which the collector current changes with V_{CE} is known as Knee voltage. The transistor always operated in the region above Knee voltage, I_C is always constant

TABULAR FORMS:

Input Characteristics:

S.No	V_{CE} = 0 V		V_{CE} = 5 V		V_{CE} = 10 V	
	V_{BE}(v)	I_B(µA)	V_{BE}(v))	I_B(µA)	V_{BE}(v))	I_B(µA)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

and is approximately equal to I_B. The current amplification factor of CE configuration is given by

$\beta = \Delta I_C / \Delta I_B$

The two sets of characteristics are necessary to describe the behavior of the CE configuration one for input or base emitter circuit and other for the output or collector emitter circuit.

Input characteristics

In input characteristics the emitter base junction forward biased by a very small voltage V_{BB} where as collector base junction reverse biased by a very large voltage V_{CC}. The input characteristics are a plot of input current I_B Verses the input voltage V_{BE} for a range of values of output voltage V_{CE}. The following important points can be observed from these characteristic curves.

The characteristics resemble that of CB configuration.

1. Input resistance is high as I_B increases less rapidly with V_{BE}
2. The input resistance of the transistor is the ratio of change in base emitter voltage ΔV_{BE} to change in base current ΔI_B at constant collector emitter voltage V_{CE} i.e... Input resistance or input impedance

$h_{ie} = \Delta V_{BE} / \Delta I_B$ at V_{CE} constant.

Output characteristics

A set of output characteristics or collector characteristics are a plot of output current I_C Verses output voltage V_{CE} for a range of values of input current I_B .The followingimportant points can be observed from these characteristic curves. The transistor always operates in the active region. i.e. the collector current I_C increases with V_{CE} very slowly. For low values of the V_{CE} the I_C increases rapidly with a small increase in V_{CE} .The transistor is said to be working in saturation region. Output resistance is the ratio of change of collector emitter voltage ΔV_{CE} ,to change in collector current ΔI_C with constant I_B. Output resistance or Output impedance $h_{oe} = \Delta V_{CE} / \Delta I_C$ at I_B constant

Input Impedance $h_{ie} = \Delta V_{BE} / \Delta I_B$ at V_{CE} constantOutput impedance $h_{oe} = \Delta I_C / \Delta V_{CE}$ at I_B constant Reverse Transfer Voltage Gain $h_{re} = \Delta V_{BE} / \Delta V_{CE}$ at I_B constant Forward Transfer Current Gain $h_{fe} = \Delta I_C / \Delta I_B$ at constant V_{CE}

APPLICATIONS:

Common Emitter configuration can be used

1. General purpose amplifier designs.
2. Digital (large signal) designs as switching stage.

Output Characteristics

S.No	I_B= 40 (µA)		I_B = 50(µA)		I_B = 60(µA)	
	V_{CE} (V)	I_C(mA)	V_{CE} (V)	I_C(mA)	V_{CE} (V)	I_C(mA)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

PROCEDURE:

Input characteristics

1. Connect the circuit as per the circuit diagram.
2. For plotting the input characteristics, the output voltage V_{CE} is kept constant at 0Vand fordifferent values of V_{BE} . Note down the values of I_B.
3. Repeat the above step by keeping V_{CE} at 5V and 10V.
4. Tabulate all the readings.
5. plot the graph between V_{BE} and I_B for constant V_{CE}

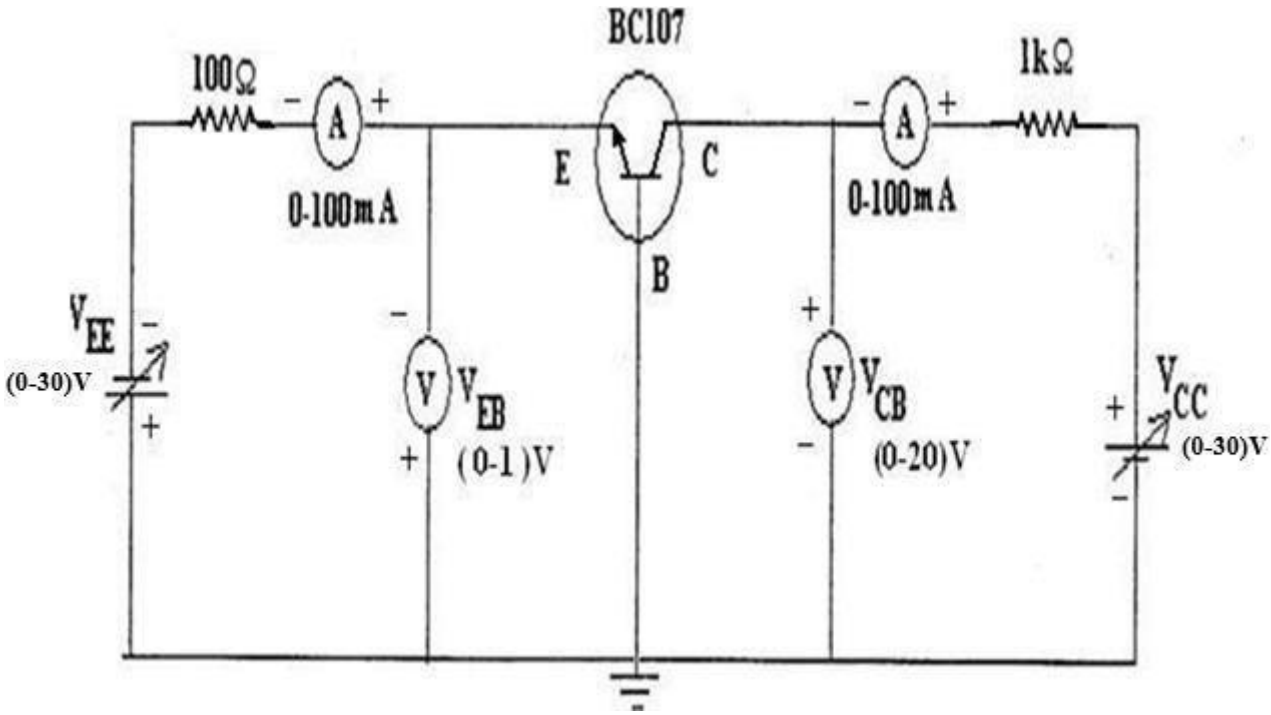
Output characteristics

2. Connections are made as per the circuit diagram.
3. For plotting the output characteristics, the input I_B is kept constant at 5μA and for differentvaluesof V_{CE}, note down the values of I_C.
4. Repeat the above step for the values of I_B at 5μA and 10μA,
5. Tabulate all the readings.
6. plot the graph between V_{CE} and I_C for constant I_B.

RESULT:

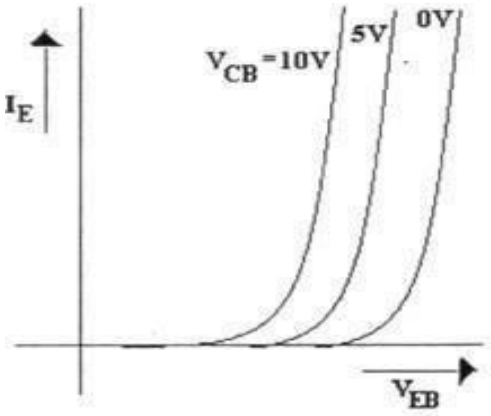
INPUT & OUTPUT CHARACTERISTICS OF BJT IN CB CONFIGURATION

CIRCUIT DIAGRAM:

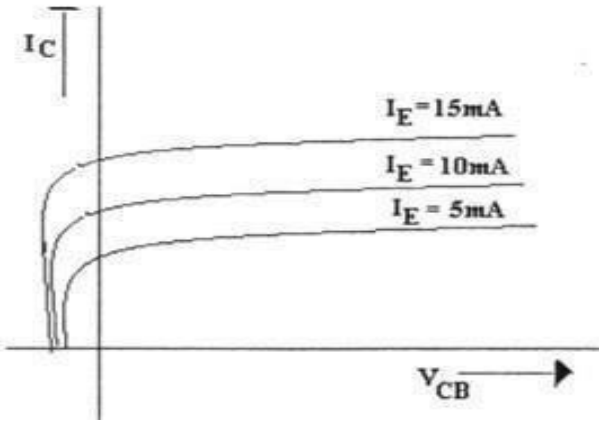


MODEL GRAPHS:

INPUT CHARACTERISTICS



OUTPUT CHARACTERISTICS



INPUT & OUTPUT CHARACTERISTICS OF BJT IN CB CONFIGURATION

AIM: To observe the input and output characteristics of a transistor in common Base Configuration.

APPARATUS REQUIRED:

S.NO	EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
1.	Transistor	BC107	1
2.	Resistor	100Ω	1
		1KΩ	1
3.	Bread Board	-	1
4.	DC Voltmeter	0-1V	1
		0-20V	1
5.	DC Ammeters	0-100mA	2
6.	Regulated power supply	0-30V	2
7.	Connecting wires	-	Few
8.	Multimeter	-	1

THEORY:

A transistor is a three terminal active device. The terminals are emitter, base and collector. In CB configuration, the base is common to both input (emitter) and output (collector). For normal operation, the E-B junction is forward biased and C-B junction is reverse biased.

In CB configuration, I_E is +ve, I_C is –ve and I_B is –ve.

So, $V_{EB}=f1(V_{CB},I_E)$ and

$I_C=f2\ (V_{CB},I_B)$

With an increasing the reverse collector voltage, the space-charge width at the output junction increases and the effective base width 'W' decreases. This phenomenon is known as "Early effect". Then,there will be less chance for recombination within the base region. With increase of charge gradient within the base region, the current of minority carriers injected across the emitter junction increases. The currentamplification factor of CB configuration is given by, $\alpha = \Delta I_C / \Delta I_E$

TABULAR FORMS:

INPUT CHARACTERISTICS

S.No	V_{CB} =0 V		V_{CB} =5 V		V_{CB} =10V	
	V_{EB}(v)	I_E(μA)	V_{EB}(v)	I_E(μA)	V_{EB}(v)	I_E(μA)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

OUTPUT CHARACTERISTICS

S.No	I_E =5mA		I_E =10mA		I_E =20mA	
	V_{CB}(V)	I_C(mA)	V_{CB}(V)	I_C (mA)	V_{CB}(V)	I_C (mA)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

PROCEDURE:

Input characteristics

1. Connections are made as per the circuit diagram.
2. For plotting the input characteristics, the output voltage V_{CE} is kept constant at 0V and for different values of V_{BE}, note down the values of I_E.
3. Repeat the above step keeping V_{CB} at 5V and 10V.All the readings are tabulated.
4. A graph is drawn between V_{EB} and I_E for constant V_{CB}.

Output characteristics

1. Connections are made as per the circuit diagram.
2. For plotting the output characteristics, the input I_E is kept constant at 5mA and fordifferent values of V_{CB}, note down the values of I_C.
3. Repeat the above step for the values of I_E at 10mA and 20mA, all the readings aretabulated.
4. A graph is drawn between V_{CB}(V)and I_C for constant I_E

APPLICATIONS:

Common Base used in

1. At high frequencies where low source resistance is common.
2. It is used for impedance matching in circuits with very low output resistances to those with a high input resistance.

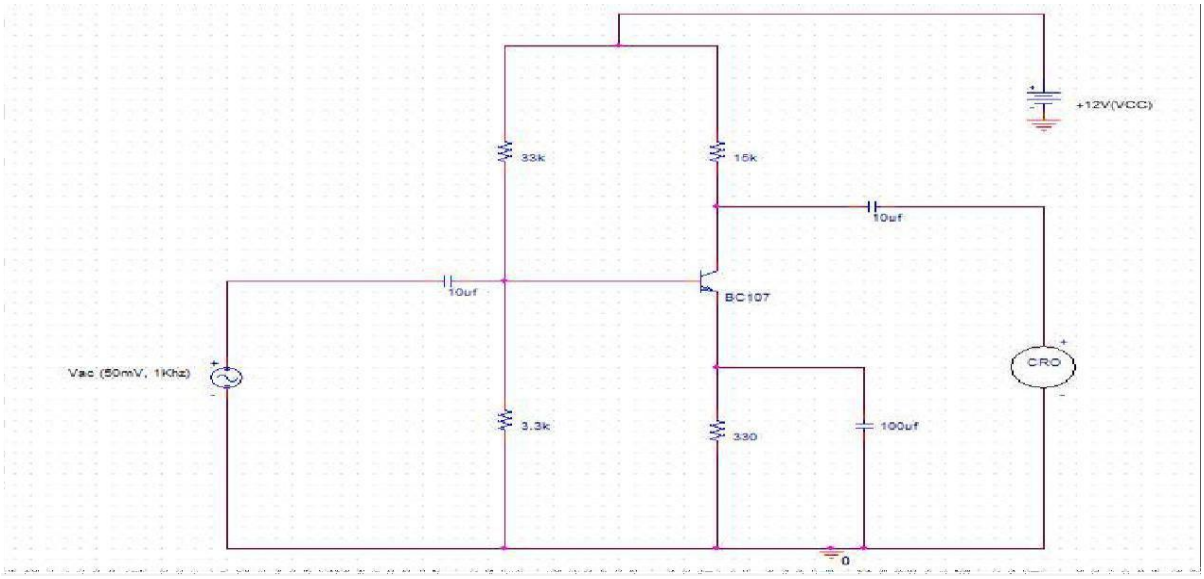
PRECAUTIONS:

1. The supply voltages should not exceed the rating of the transistor.
2. Meters should be connected properly according to their polarities.

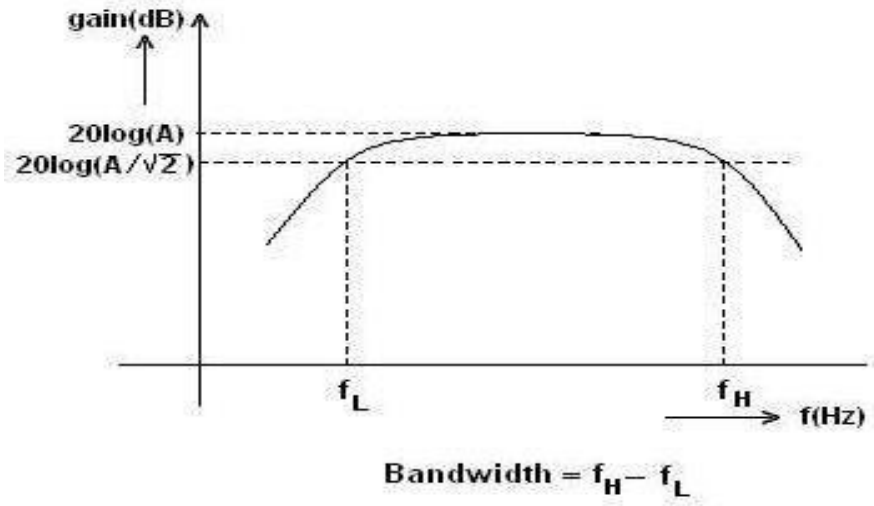
RESULT:

COMMON EMITTER AMPLIFIER
(CE AMPLIFIER)

CIRCUIT DIAGRAM:



MODEL GRAPHS:



EXPERIMENT NO: 11

CE AMPLIFIER

AIM: To Observe the frequency response of a common emitter amplifier using hardware.

APPARATUS:

EQUIPMENT/COMPONENT	SPECIFICATIONS	QUANTITY
Transistor	BC107	1
Resistor	1.5KΩ	1
	33KΩ	1
	330Ω	1
	3.3KΩ	1
Capacitor	10µF	2
	100µF	1
FunctionGenerator		1
BreadBoard		1
CRO		1

THEORY:

The CE amplifier provides high gain & wide frequency response. The emitter terminal is common to both input & output circuits and is grounded. The emitter-base circuit is forward biased. The collector current is controlled by the base current rather than emitter current. The input signal is applied to base terminal of the transistor and amplifier output is taken across collector terminal. A very small change in base current produces a much larger change in collector current. When +VE half-cycle is fed to the input circuit, it increases the forward bias of the circuit which causes the collector current to increase, it decreases the voltage more VE. Thus when input cycle varies through a -VE half-cycle, decreases the forward bias of the circuit, which causes the collector current to increases thus the output signal is common emitter amplifier is in out of phase with the input signal.

The gain of the amplifier increases as the frequency increases from zero till it becomes maximum at lower cut-off frequency and remains constant till higher cut-off frequency and then it falls again as the frequency increases. At low frequencies the reactance of coupling capacitor CC is quite high and hence very small part of signal will pass through from one stage to the next stage. At high frequencies the reactance of inter electrode capacitance is very small and behaves as a short circuit.

OBSERVATIONTABLE:

Input Voltage (Vin)= mV

S.No	Input Frequency (Hz)	Output Voltage (Vo)	Gain(Av)=Vo/Vi	Gain in dB= 20log10(Vo/Vi)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				

This increases the loading effect on next stage and service to reduce the voltage gain due to these reasons the voltage gain drops at high frequencies. At mid frequencies the effect of coupling capacitors is negligible and acts like short circuit, whereas inter electrode capacitors acts like open circuit. So, the circuit becomes resistive at mid frequencies and the voltage gain remains constant during this range.

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Apply input of 50Mv peak to peak and 1Khz frequency using function generator.
3. Measure the output voltage Vo peak to peak for various input frequencies ranging from 10Hz to 1Mhz.
4. Tabulate the readings in tabular column.
5. The voltage gain can be calculated by using expression Av= Vo/Vi
6. The voltage gain in dB can be calculated by using expression $Av=20\log_{10}(Vo/Vi)$.
7. A semi log graph drawn by taking frequency on x-axis and gain in dB in y-axis.
8. The bandwidth of amplifier is calculated from the graph using expression BW= fH-fL
9. The bandwidth product of amplifier is calculated using the expression gain bandwidth product equal to 3dB mid band gain * bandwidth.

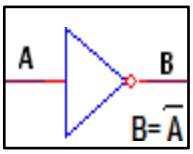
PRECAUTIONS:

1. Connect the circuit without errors.
2. Use the circuit components with proper ratings.
3. Ground the circuit properly.

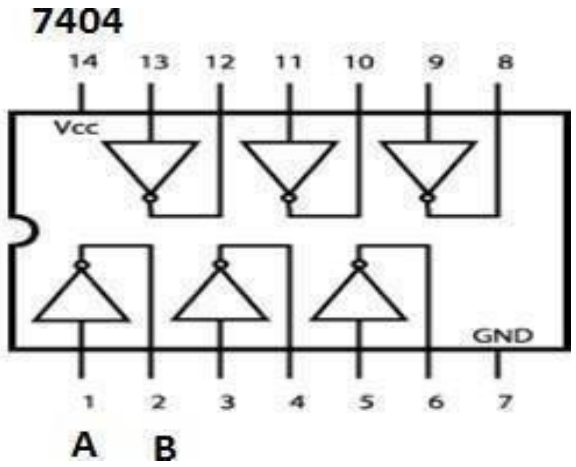
RESULT:

VERIFICATION OF LOGIC GATES

NOT gate:

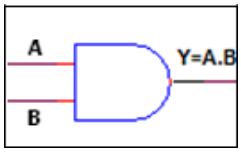


Truth Table

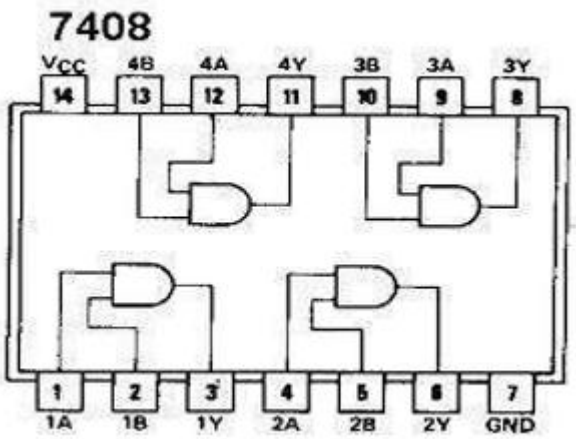


INPUTS	OUTPUT	VERIFIED OUTPUT
A	B=Ā	ON/OFF
0		
1		

AND gate:



Truth Table



INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	Y=A.B	ON/OFF
0	0		
0	1		
1	0		
1	1		

EXPERIMENT NO: 12

VERIFICATION OF LOGIC GATES

AIM: To verify the behavior (or truth table) of logic gates using IC s.

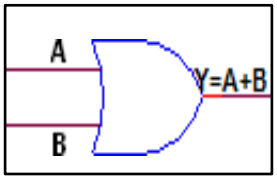
APPARATUS REQUIRED:

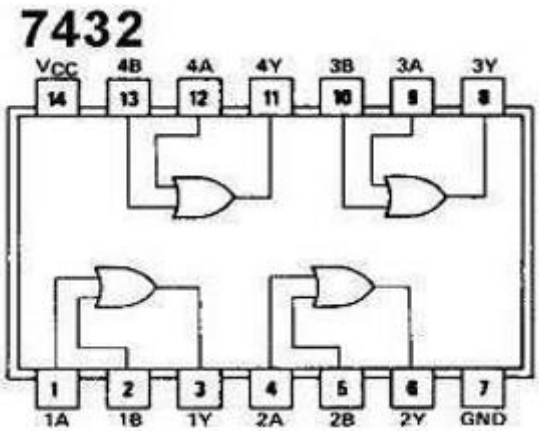
S.NO	EQUIPMENT /COMPONENT NAME	QUANTITY
1.	Component development system (bread board trainer)	1
2.	IC 7400 (NAND)	1
3.	IC 7402 (NOR)	1
4.	IC 7404 (NOT)	1
5.	IC 7408 (AND)	1
6.	IC 7432 (OR)	1
7.	IC 7486 (EX-OR)	1
8.	IC 74266 (EX-NOR)	1
9.	Patch chords & Connecting Wires	As per Required

THEORY:

- The NAND gate performs the logical AND-INVERT of two variables A and B produces an output C.
- The NOR gate performs the logical OR-INVERT of two variables A and B produces an output C.
- The NOT gate performs the logical INVERT of input variable A and produces an output B.
- The AND gate performs the logical AND of two variables A and B produces an output C.
- The OR gate performs the logical OR of two variables A and B produces an output C.
- The EX-OR gate performs the logical EX-OR of two variables A and B produces an output C.
- The EX-NOR gate performs the logical EX-NOR of two variables A and B produces an output C.

OR gate:

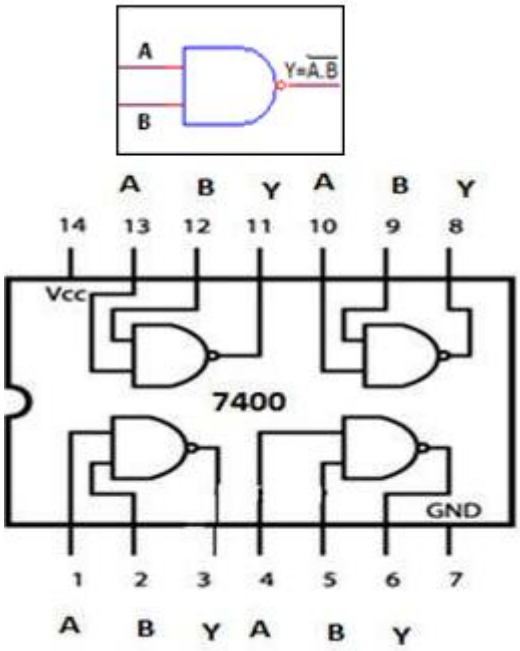




Truth Table

INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	Y=A+B	ON/OFF
0	0		
0	1		
1	0		
1	1		

NAND gate:



Truth Table

INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	$Y=\overline{A.B}$	ON/OFF
0	0		
0	1		
1	0		
1	1		

PROCEDURE:

❖ NOT Gate:

1. Take the component development system (bread board) and insert the IC 7404 at appropriate position.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic input and take the output for different inputs.

❖ AND Gate:

1. Insert IC 7408 at appropriate position on component development system.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic inputs and take the output for different combinations of inputs.

❖ OR Gate:

1. Insert IC 7432 at appropriate position on component development system.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic inputs and take the output for different combinations of inputs.

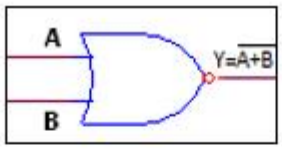
❖ NAND Gate

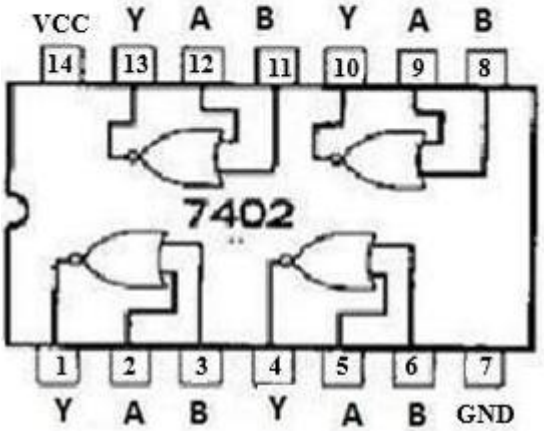
1. Take the component development system (bread board) and insert the IC 7400 at appropriate position.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic inputs and take the output for different combinations of inputs.

❖ NOR Gate:

1. Insert IC 7402 at appropriate position on component development system.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic inputs and take the output for different combinations of inputs.

NOR gate:

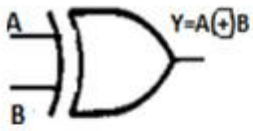


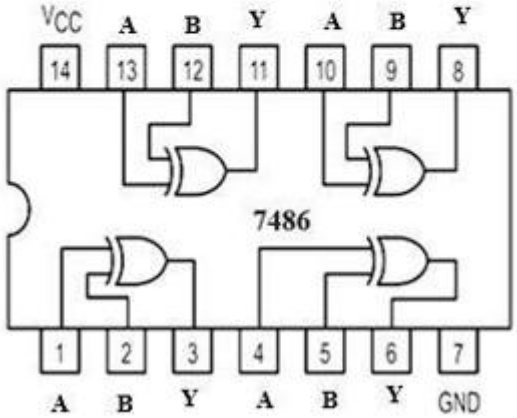


Truth Table

INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	$Y=\overline{A+B}$	ON/OFF
0	0		
0	1		
1	0		
1	1		

EX-OR gate:





Truth Table

INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	$Y=A\oplus B$	ON/OFF
0	0		
0	1		
1	0		
1	1		

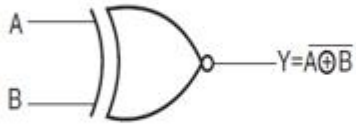
❖ EX-OR Gate:

1. Insert IC 7486 at appropriate position on component development system.
2. Connect pin No 14 to 5V and pin No.7 to GND.
3. Apply the logic inputs and take the output for different combinations of inputs.

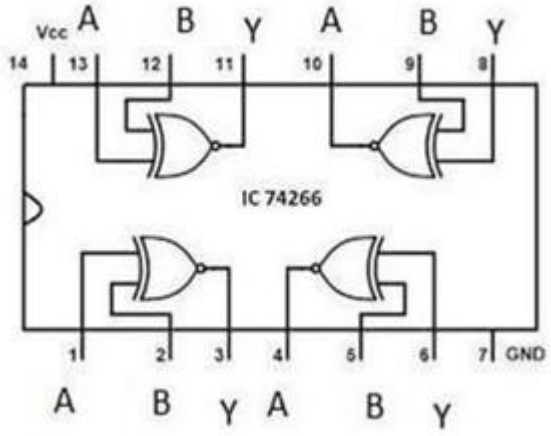
❖ EX-NOR Gate:

4. Insert IC 74266 at appropriate position on component development system.
5. Connect pin No 14 to 5V and pin No.7 to GND.
6. Apply the logic inputs and take the output for different combinations of inputs.

Exclusive-NOR gate:



$$Y = (\overline{A \oplus B}) = (A.B + \overline{A}.\overline{B})$$



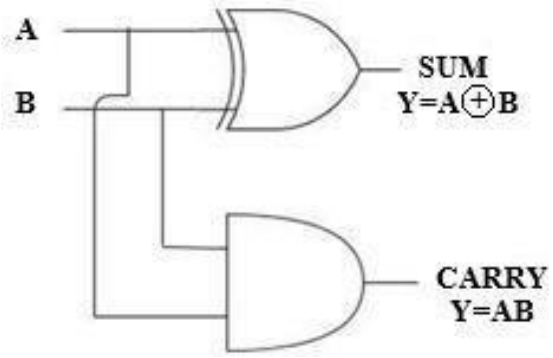
Truth Table

INPUTS		OUTPUT	VERIFIED OUTPUT
A	B	$Y=\overline{A \oplus B}$	ON/OFF
0	0		
0	1		
1	0		
1	1		

RESULT:

DESIGN AND VERIFY HALF ADDER AND FULL ADDER CIRCUITS

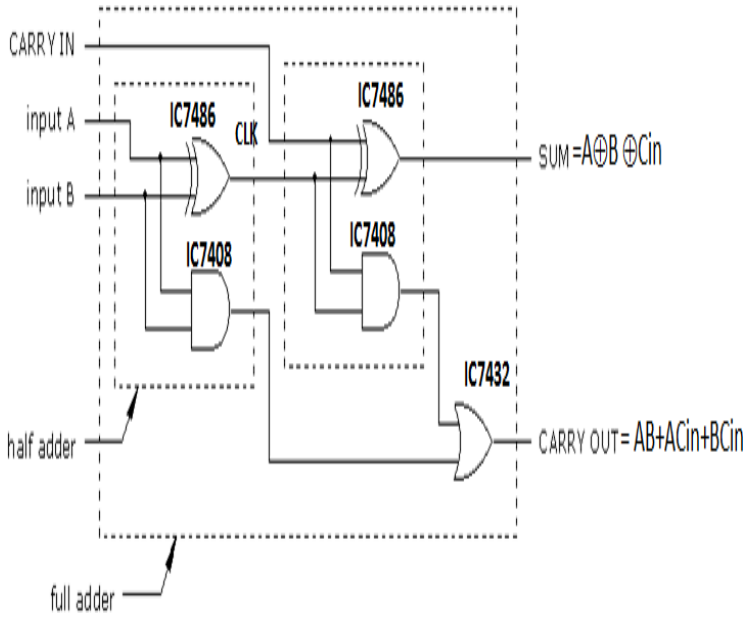
Logic Diagram of Half Adder



Truth Table

Inputs		Outputs	
A	B	Carry	Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Logic Diagram of Full Adder Using Two Half Adders



Truth Table

A	B	Cin	Cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

EXPERIMENT NO: 13

DESIGN AND VERIFY HALF ADDER AND FULL ADDER CIRCUITS

AIM: To construct Half Adder and full Adder and verify Truth Table.

APPARATUS REQUIRED:

S.NO	EQUIPMENT /COMPONENT NAME	QUANTITY
1.	Component development system (bread board trainer)	1
2.	IC 7404 (NOT)	1
3.	IC 7408 (AND)	1
4.	IC 7432 (OR)	1
5.	IC 7486 (EX-OR)	1
6.	Patch chords & Connecting Wires	As per Required

THEORY:

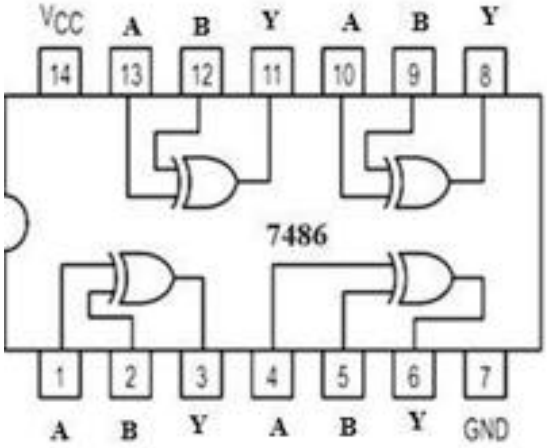
Half Adder (HA):

Half adder is the simplest of all adder circuits. Half adder is a combinational arithmetic circuit that adds two numbers and produces a sum bit (s) and carry bit (c) both as output. The addition of 2 bits is done usinga combination circuit called a Half adder. The input variables are augend and addend bits and output variables are sum & carry bits. A and B are the two input bits. let us consider two input bits A and B, then sum bit (s) is the X-OR of A and B. it is evident from the function of a half adder that it requires one X-OR gate and one AND gate for its construction.

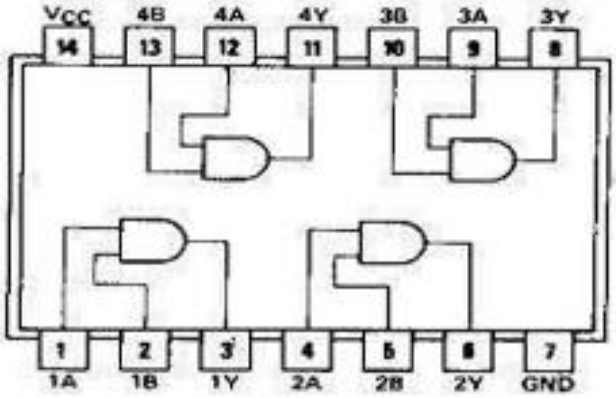
Full Adder (FA):

A full adder is a combinational circuit that forms the arithmetic sum of input; it consists of three inputs and two outputs. A full adder is useful to add three bits at a time but a half adder cannot do so. In full adder sum output will be taken from X-OR Gate, carry output will be taken from OR Gate. The full-adder has three inputs and two outputs. The first two inputs are A and B and the third input is an input carry. When a full adder logic is designed we will be able to string eight of them together to create a byte-wide adder and cascade the carry bit from one adder to the next.

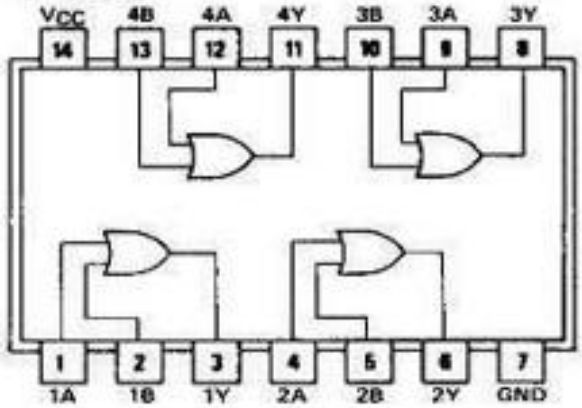
7486



7408



7432



PROCEDURE:

1. Connect the trainer kit to ac power supply.
2. Connect logic sources to the inputs of the adder.
3. Connect output from SUM and CARRY to logic indicators.
4. Apply various input combinations to the adder.
5. Observe the SUM and CARRY outputs, verify the truth table for each input/ output combination.
6. Switch off the ac power supply.

RESULT: